Learning from Examples

Topics

- ML & Agents
- Supervised Learning
- Data
- Training & Testing
- Types of supervised ML Models
- Use in AI

ML and Agents

DeepAi.org with prompt: “A happy cartoon robot with an artificial neural network for a brain on white background learning to play chess”

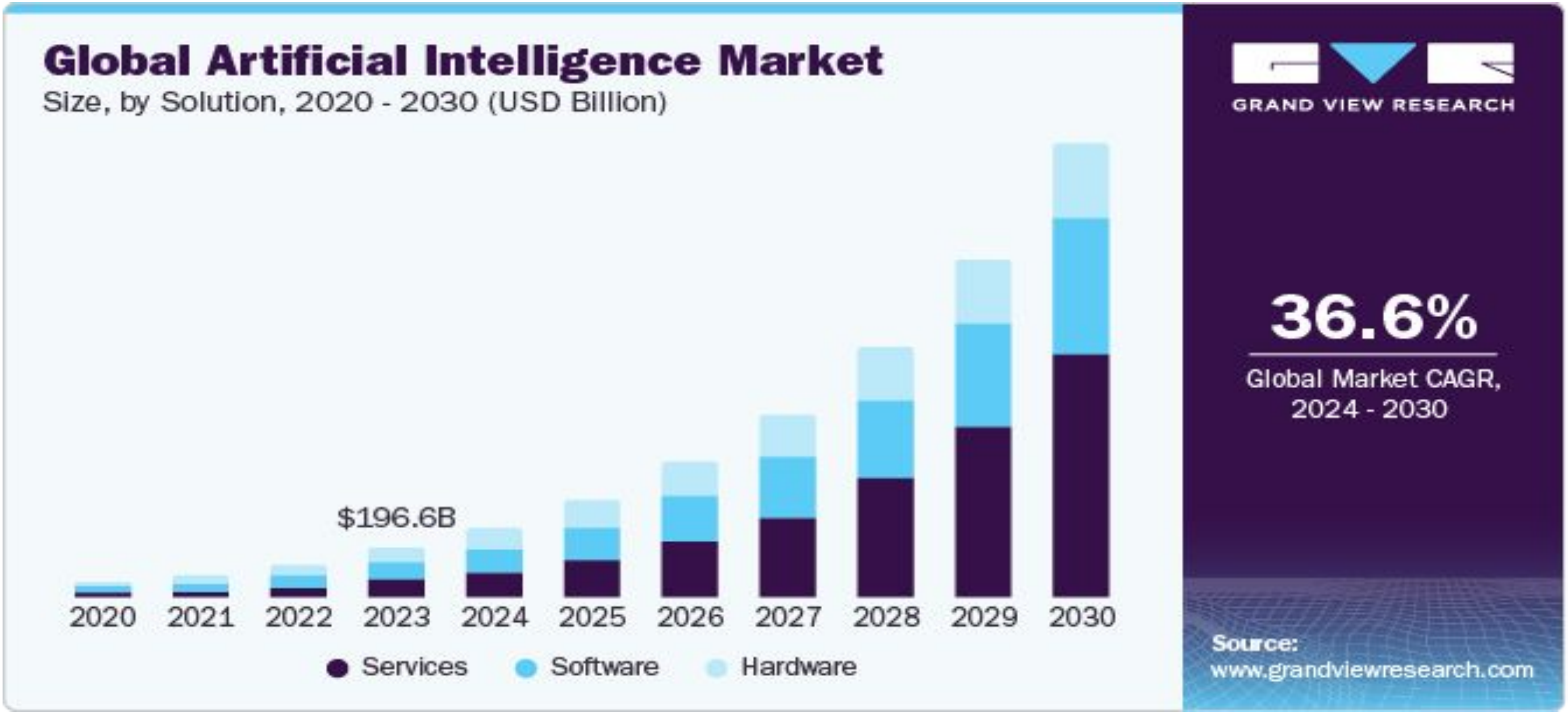
Recent Trends in AI and Machine Learning

- **Artificial Intelligence and machine learning** have been hot topics in 2023 as AI and ML technologies increasingly find their way into everything from advanced **quantum computing systems and leading-edge medical diagnostic systems to consumer electronics and “smart” personal assistants.**

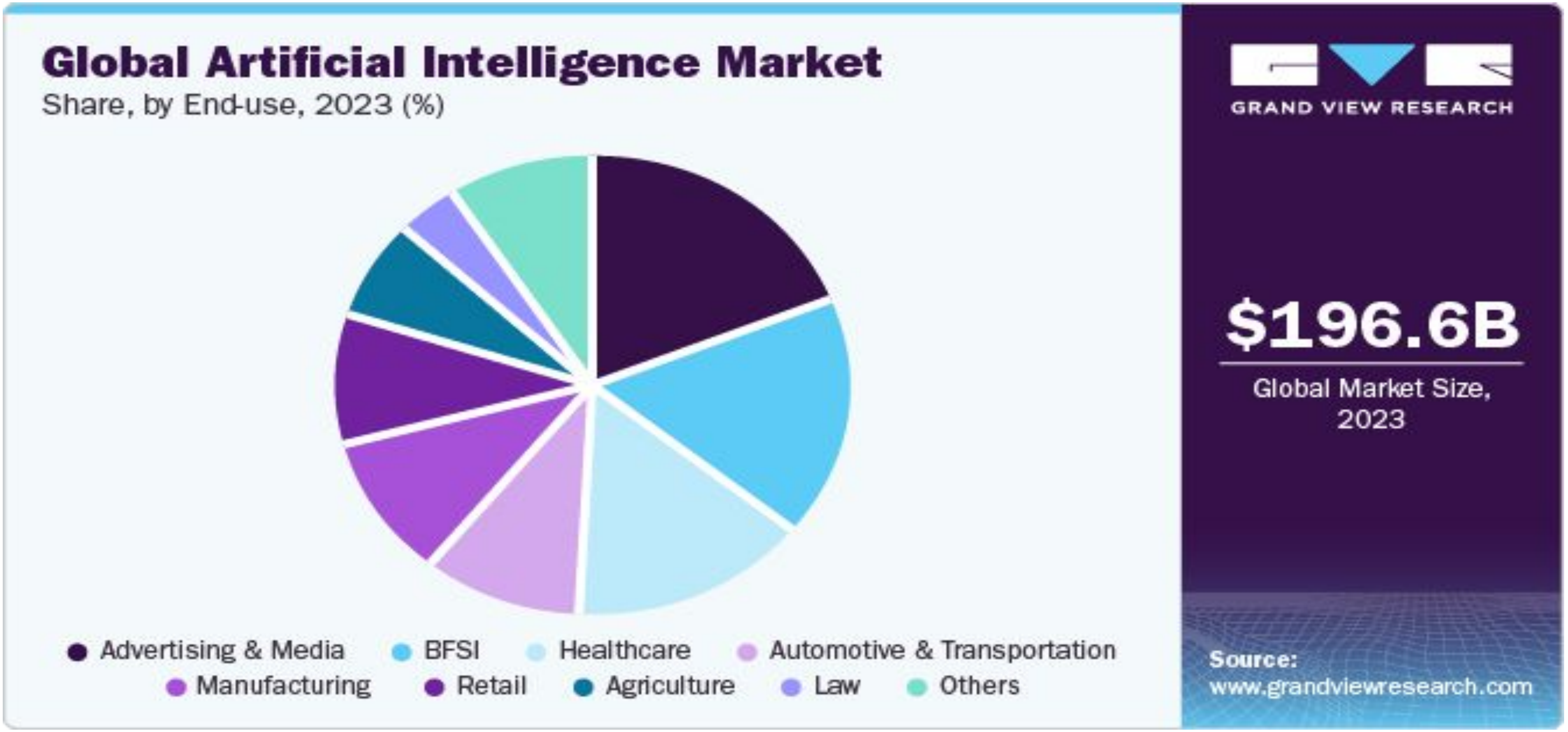
Research Scope in AI and Machine Learning

- Revenue generated by **AI hardware, software and services** is **expected to reach $156.5 billion worldwide this year,** according to market researcher.

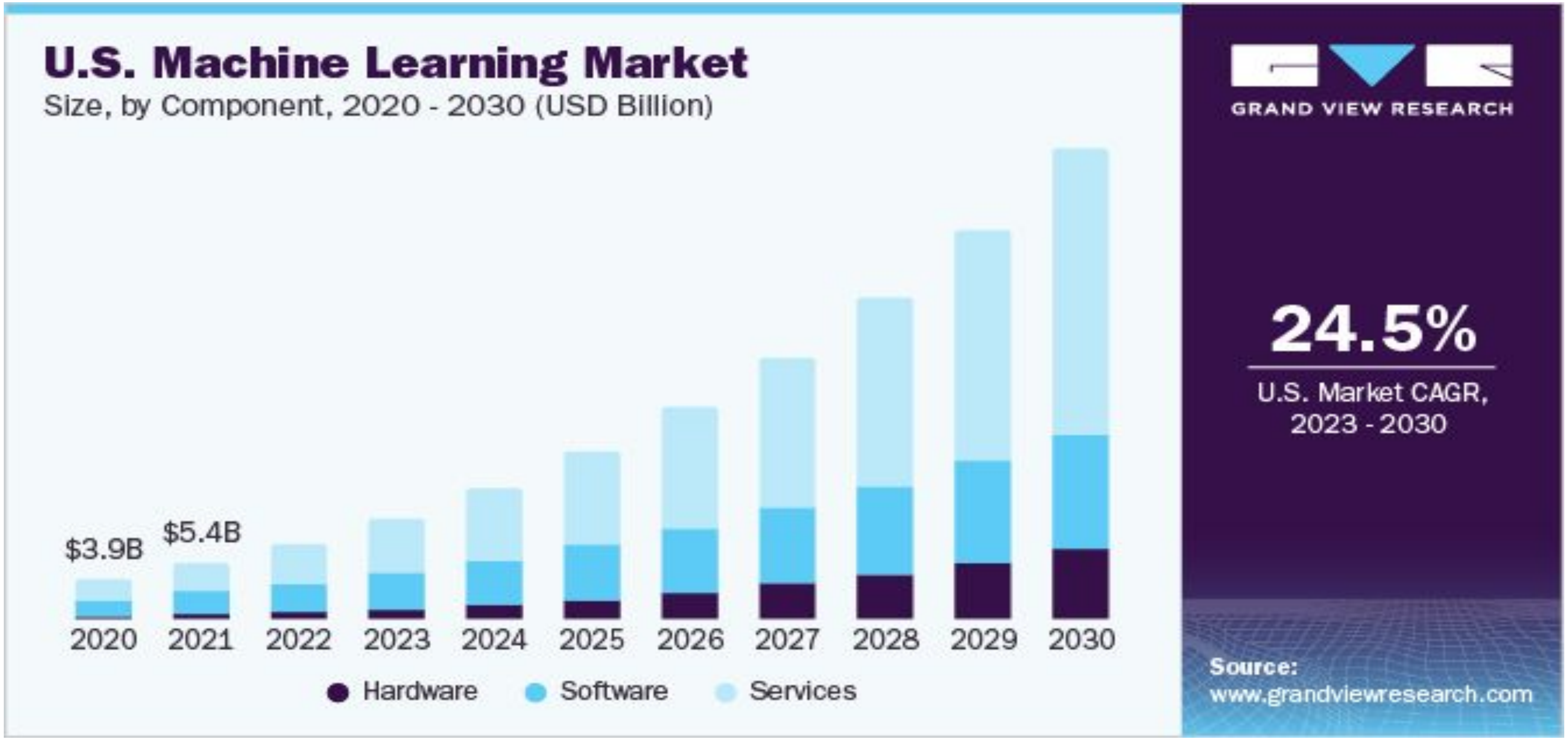
The global AI market is expected to reach $1.81 trillion by 2030



Artificial Intelligence Market Size, Share, Growth Report 2030



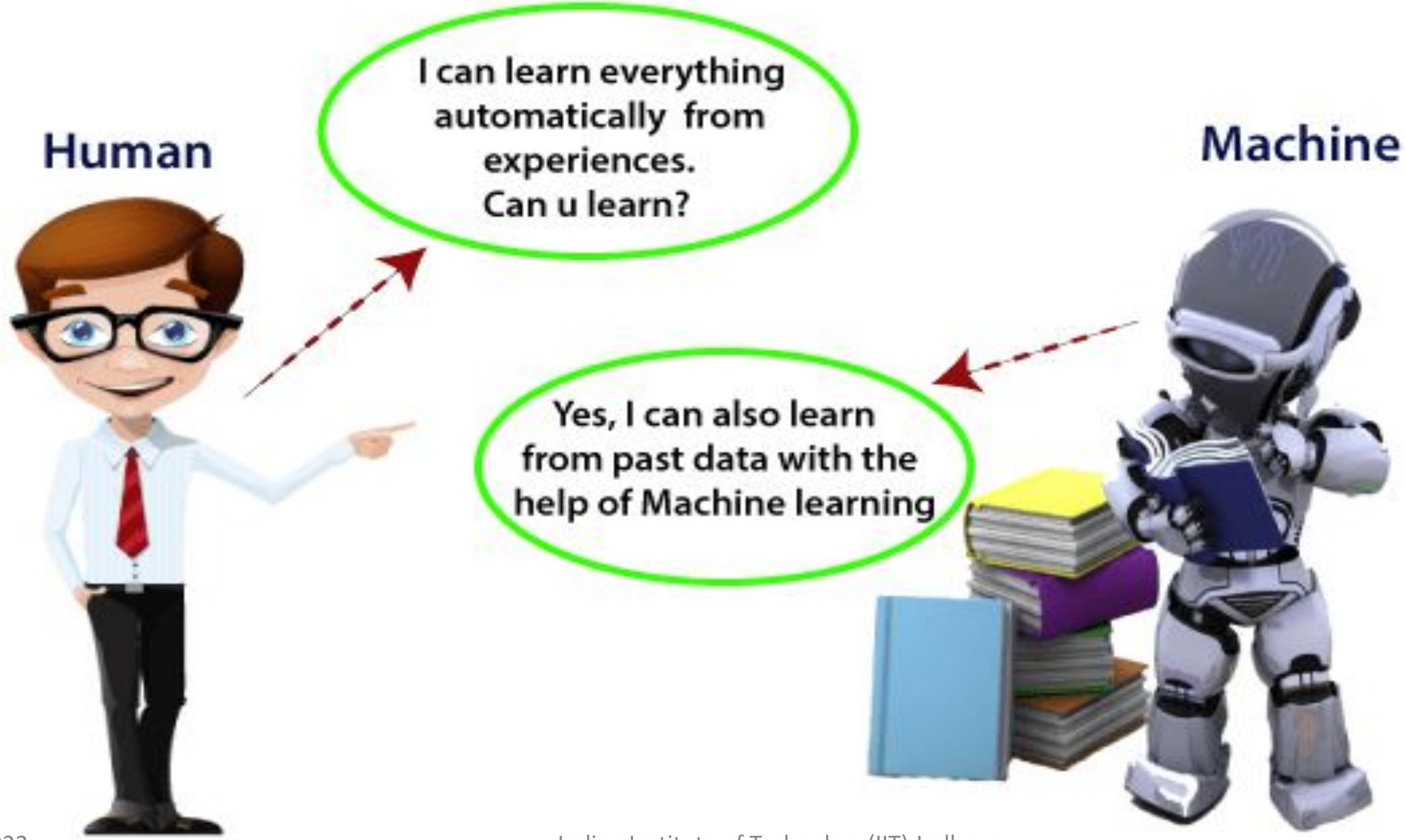
Machine Learning Market Size, Share & Growth Report, 2030



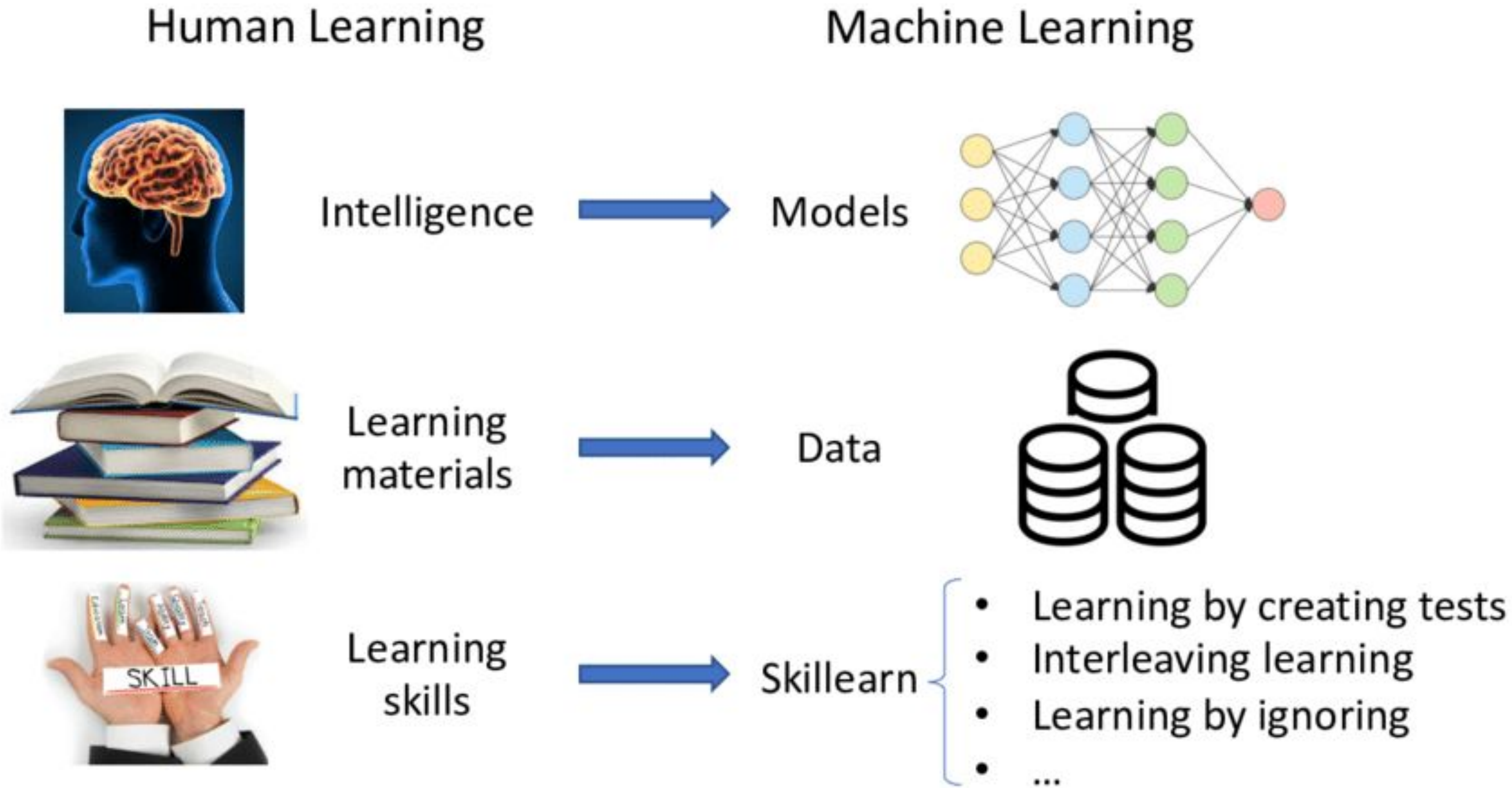
Machine Learning

- Machine learning allows computers to **find hidden insights** through **iteratively learning from data**, without being explicitly programmed.
- It has revolutionized the world of **computer science** by allowing **learning with large datasets**, which enables machines to **change, re-structure and optimize algorithms** by themselves.
- Machine learning (ML) is **a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so**.
- **Machine learning algorithms** use historical data as input to predict new output values.

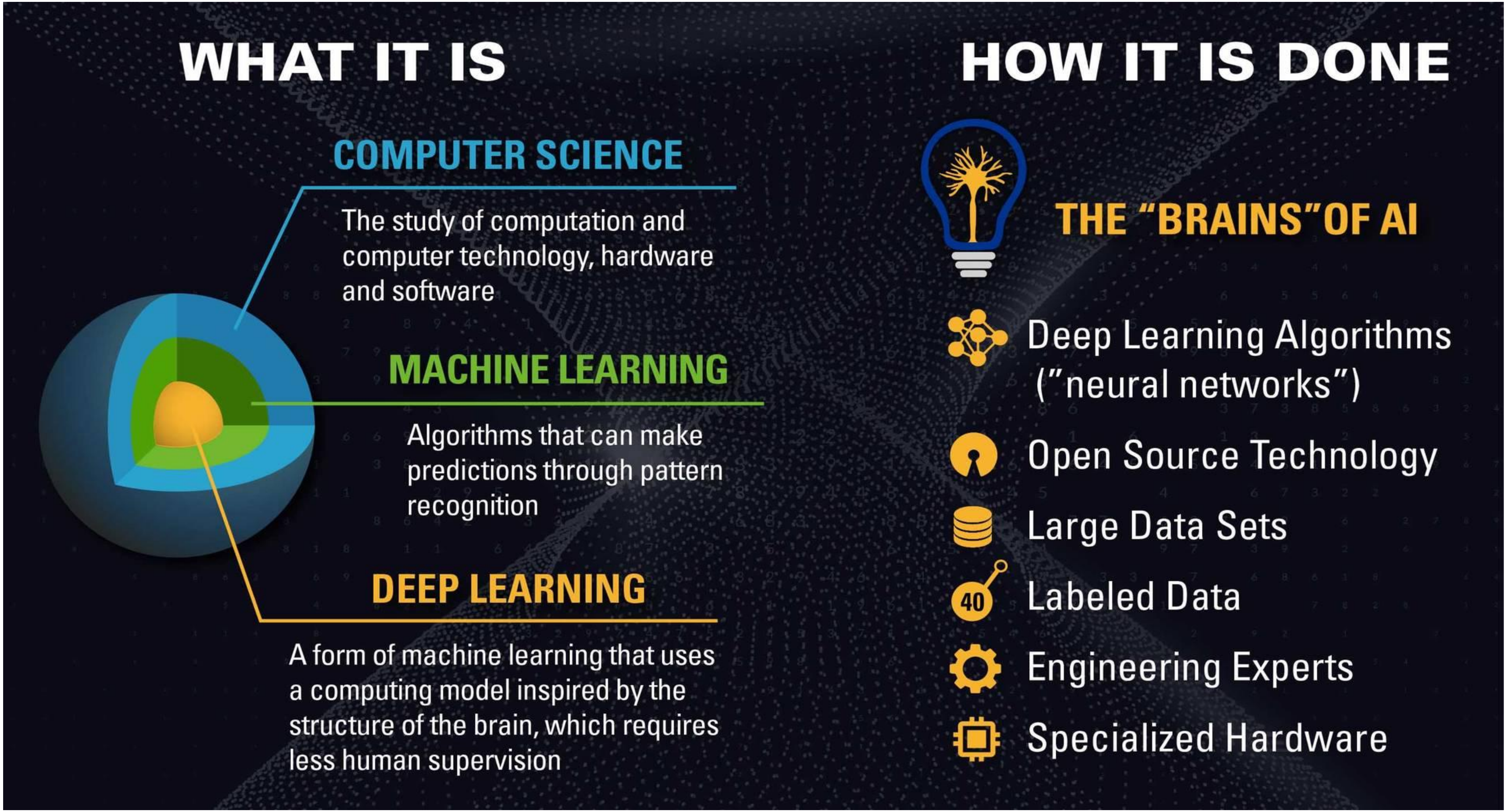
Machine Learning

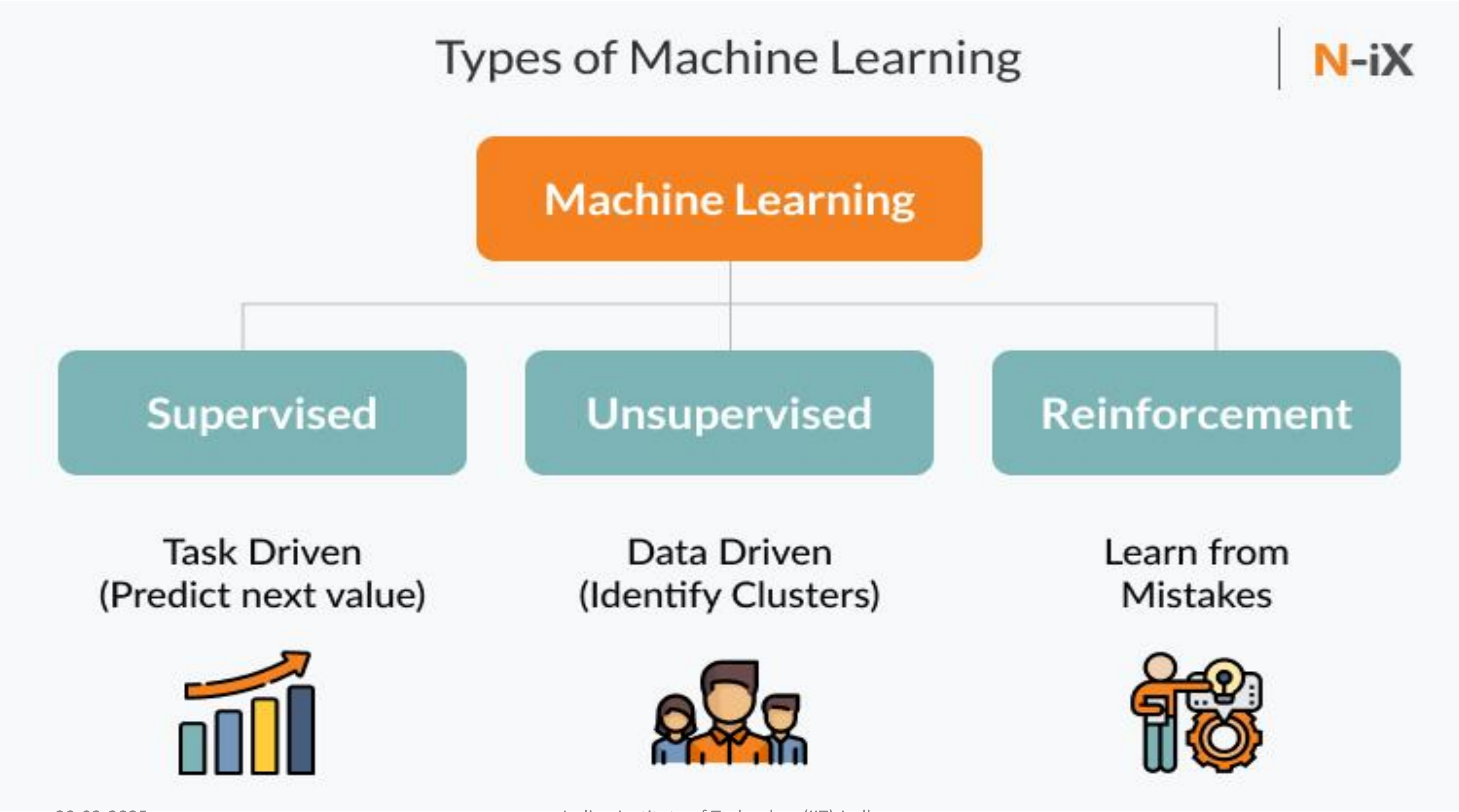


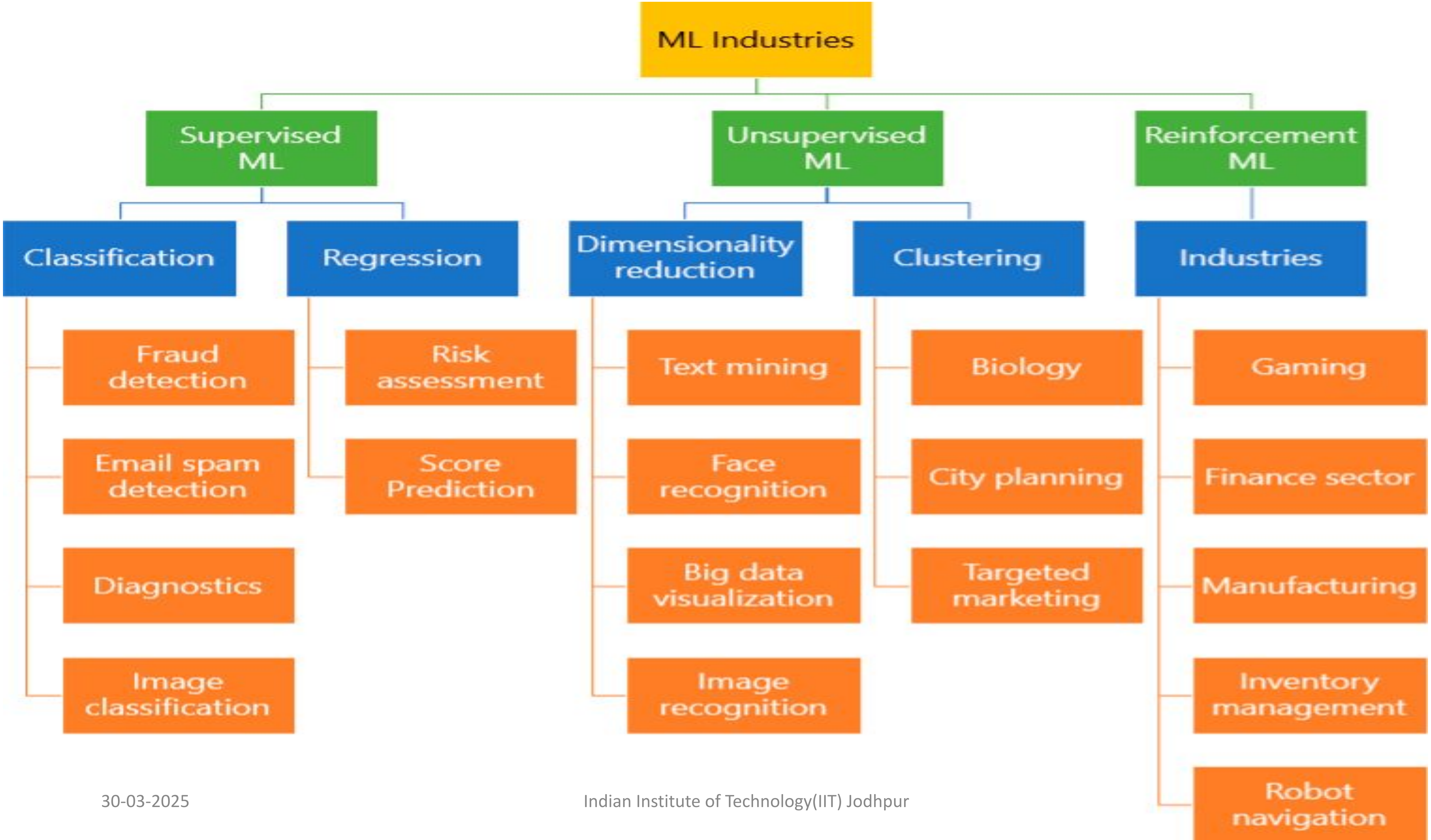
HL Vs ML



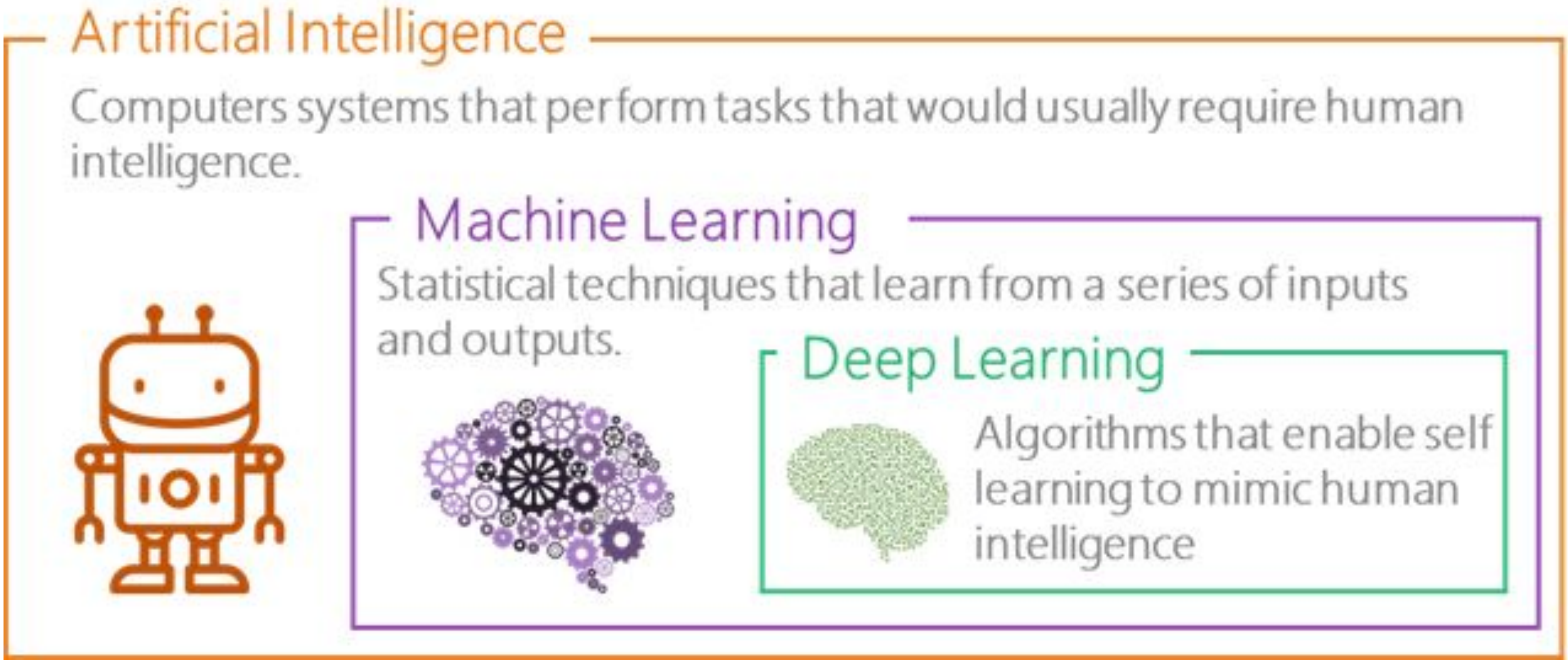
Human Learning
Machine Learning
Intelligence
Models
Learning materials
Data
SKILL
Learning skills
Skillearn
• Learning by creating tests
• Interleaving learning
• Learning by ignoring
• ...


WHAT IT IS
COMPUTER SCIENCE
The study of computation and computer technology, hardware and software
MACHINE LEARNING
Algorithms that can make predictions through pattern recognition
DEEP LEARNING
A form of machine learning that uses a computing model inspired by the structure of the brain, which requires less human supervision
HOW IT IS DONE
THE "BRAINS" OF AI
Deep Learning Algorithms ("neural networks")
Open Source Technology
Large Data Sets
40
Labeled Data
Engineering Experts
Specialized Hardware

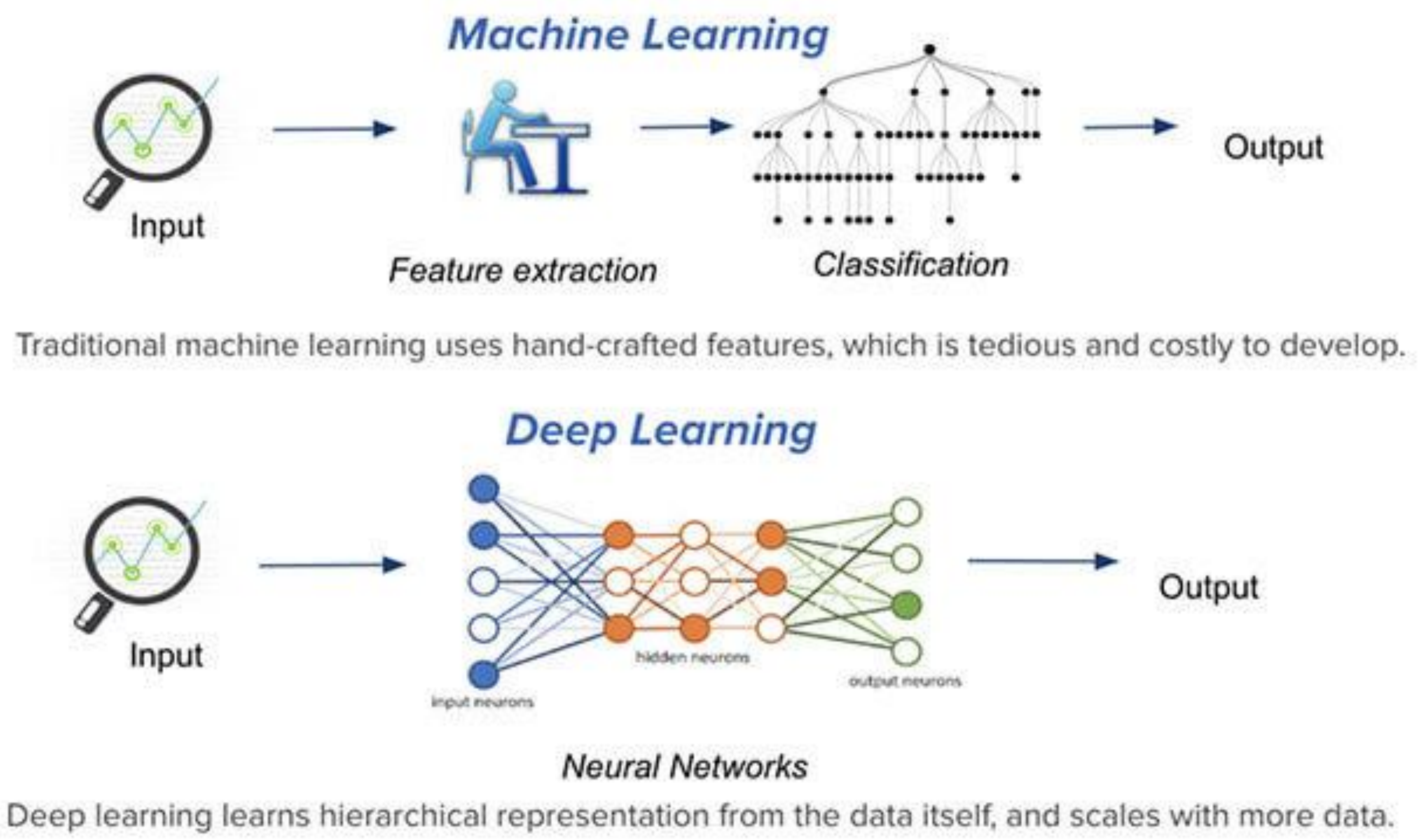

Types of Machine Learning
N-iX
Machine Learning
Supervised
Unsupervised
Reinforcement
Task Driven
(Predict next value)
Data Driven
(Identify Clusters)
Learn from
Mistakes


ML Industries
Supervised ML
Unsupervised ML
Reinforcement ML
Classification
Regression
Dimensionality reduction
Clustering
Industries
Fraud detection
Email spam detection
Diagnostics
Image classification
Risk assessment
Score Prediction
Text mining
Face recognition
Big data visualization
Image recognition
Biology
City planning
Targeted marketing
Gaming
Finance sector
Manufacturing
Inventory management
Robot navigation

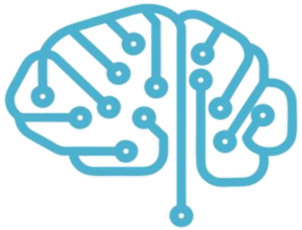
AI, ML and DL


Artificial Intelligence
Computers systems that perform tasks that would usually require human intelligence.
Machine Learning
Statistical techniques that learn from a series of inputs and outputs.
Deep Learning
Algorithms that enable self learning to mimic human intelligence

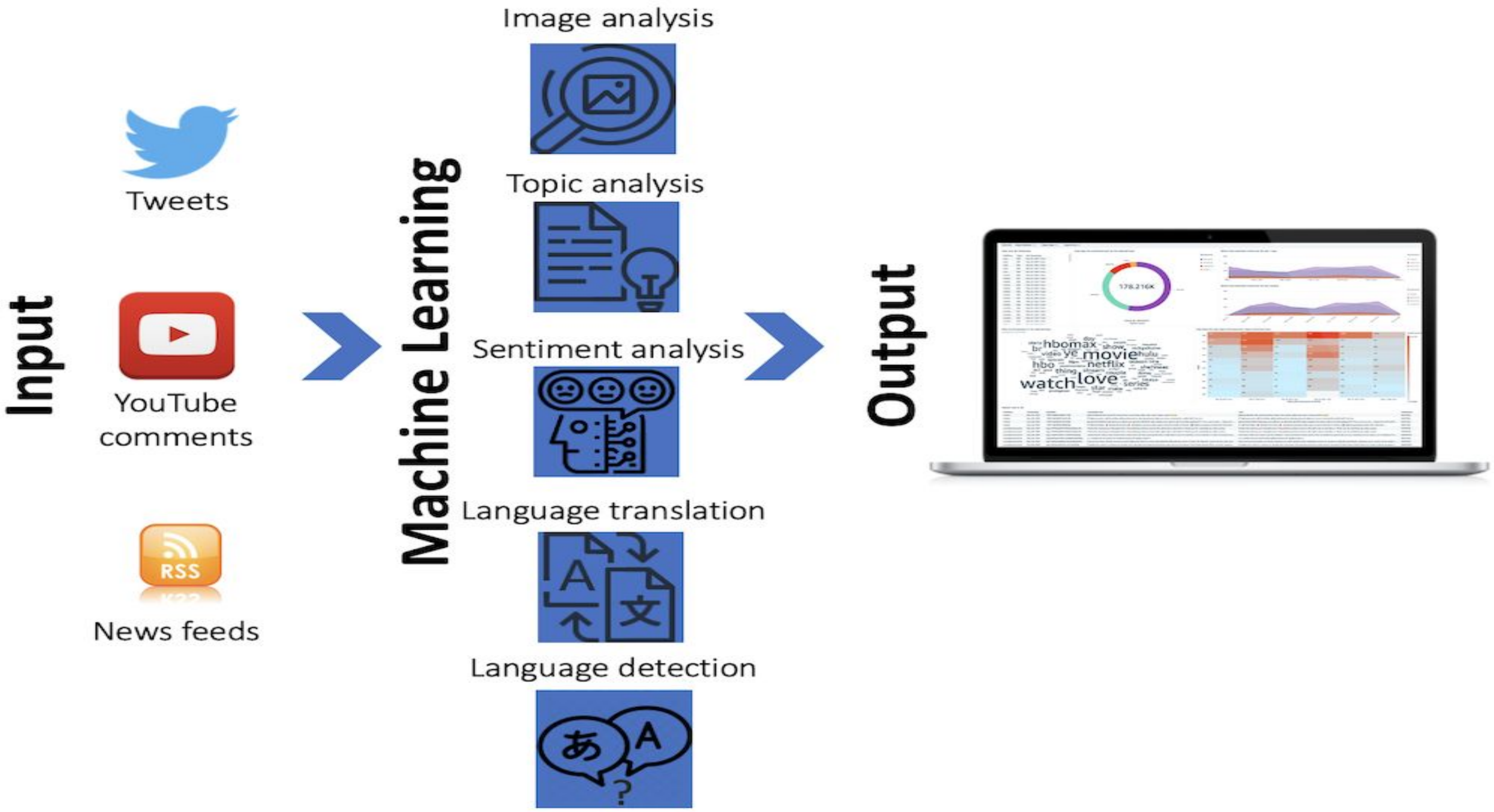
ML Vs DL


Machine Learning
Input
Feature extraction
Classification
Output
Traditional machine learning uses hand-crafted features, which is tedious and costly to develop.
Deep Learning
Input
input neurons
hidden neurons
output neurons
Output
Neural Networks
Deep learning learns hierarchical representation from the data itself, and scales with more data.

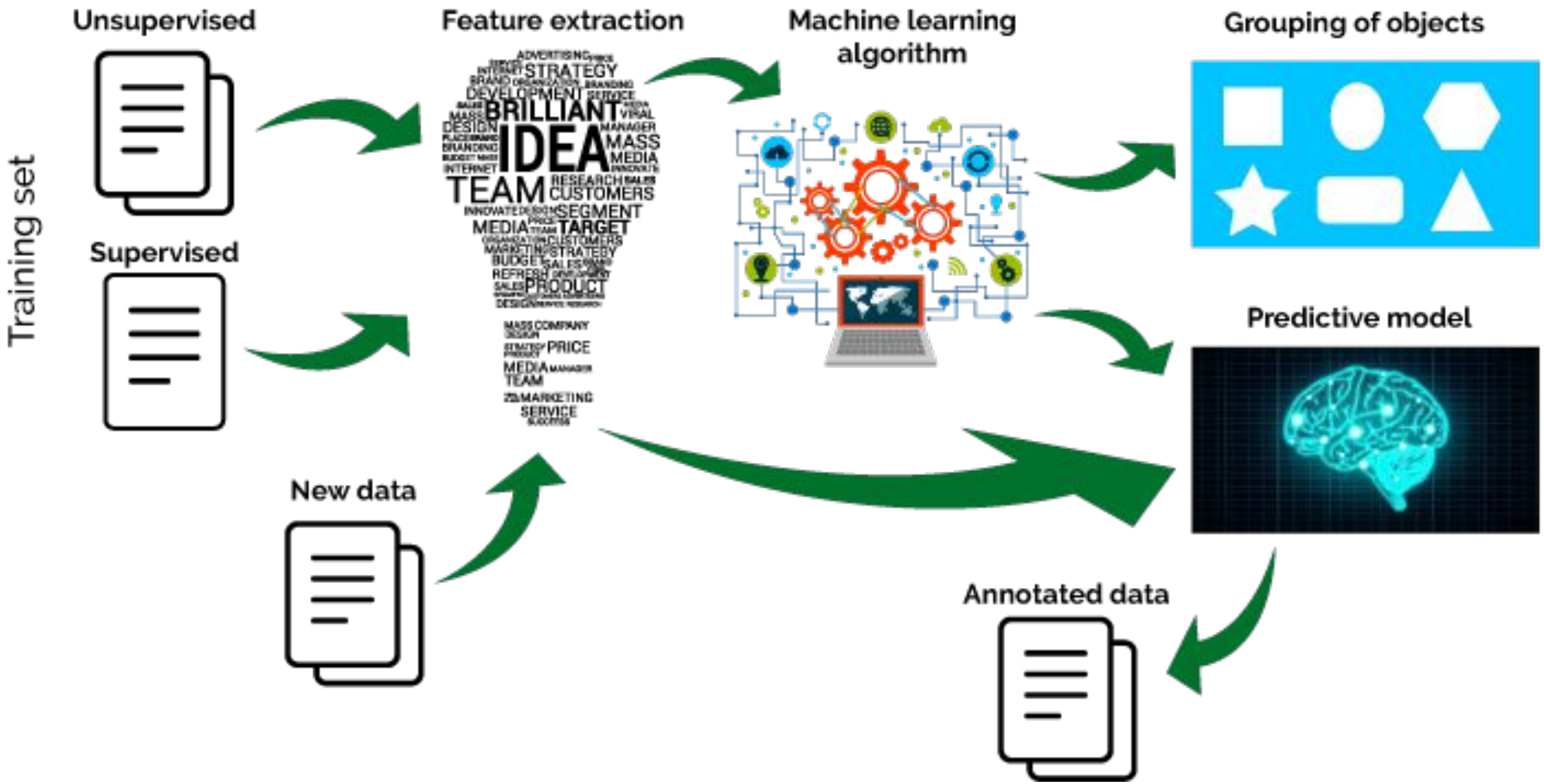
AI, ML and DL

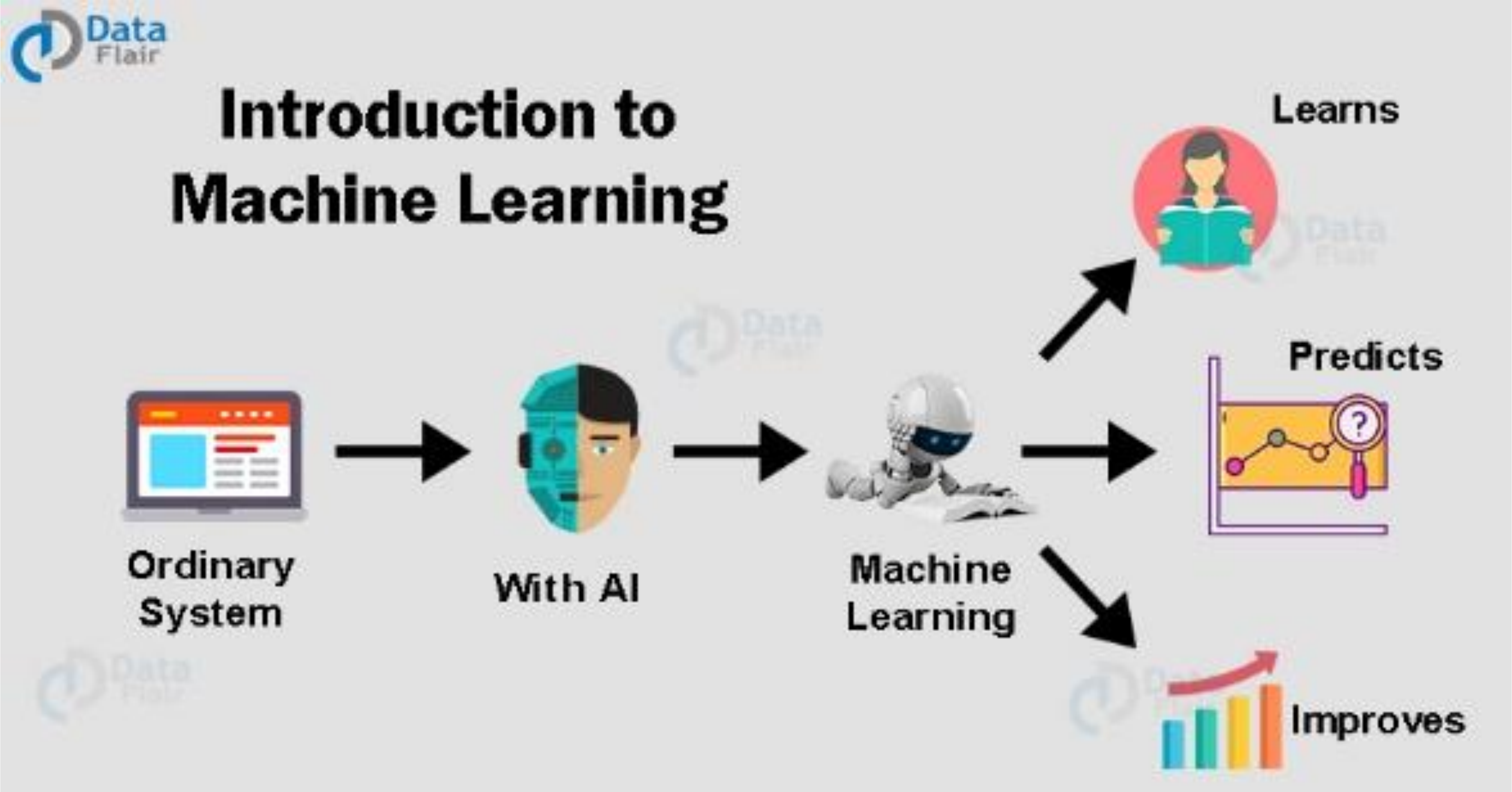
 Artificial Intelligence	 Machine Learning	 Deep Learning
Artificial intelligence originated around 1950s.	Machine learning originated around 1960s.	Deep learning originated around 1970s.
AI represents simulated intelligence in machines.	Machine Learning is the practice of getting machines to make decisions without being programmed.	Deep Learning is the process of using Artificial Neural Networks to solve complex problems.
AI is a subset of Data Science.	Machine learning is a subset of AI & Data Science	Deep learning is a subset of Machine learning, AI & Data Science.
Aim is to build machines which are capable of thinking like humans.	Aim is to make machines learn through data so that they can solve problems.	Aim is to build neural networks that automatically discover patterns for feature detection.

How It Works


Input
Tweets
YouTube comments
News feeds
Machine Learning
Image analysis
Topic analysis
Sentiment analysis
Language translation
Language detection
Output

Machine Learning




Data Flair
Introduction to
Machine Learning
Learns
Predicts
Ordinary System
With AI
Machine Learning
Improves

Machine Learning Approaches

- Existing machine learning methods can be divided into **three categories, namely, supervised learning, unsupervised learning, and reinforcement learning.**

- Other learning schemes, such as **semi-supervised learning, online learning, and transfer learning**, can be viewed as variants of these three basic types.

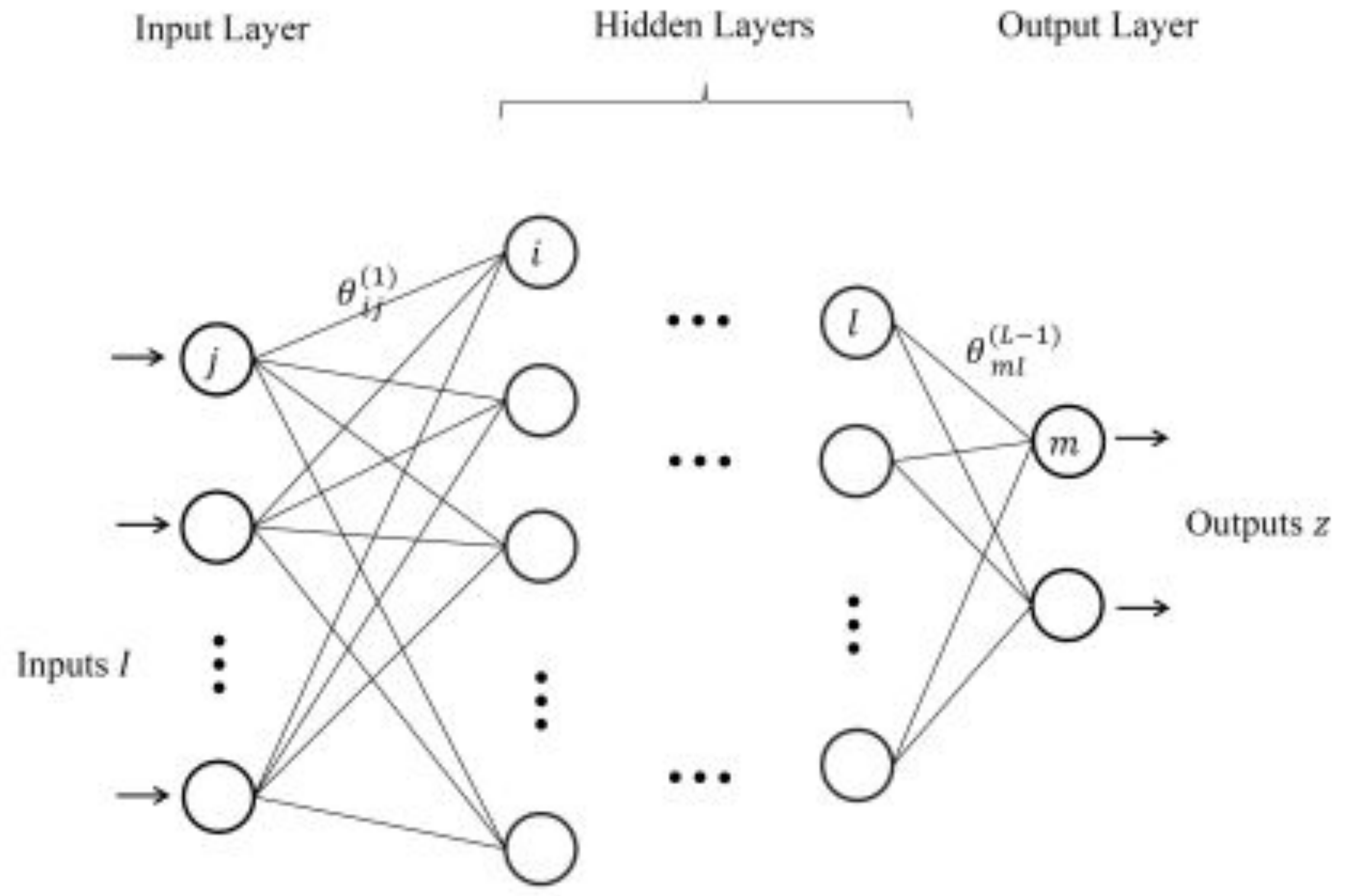
Machine Learning Stages

- In general, machine learning involves two stages, i.e., **training and testing.**

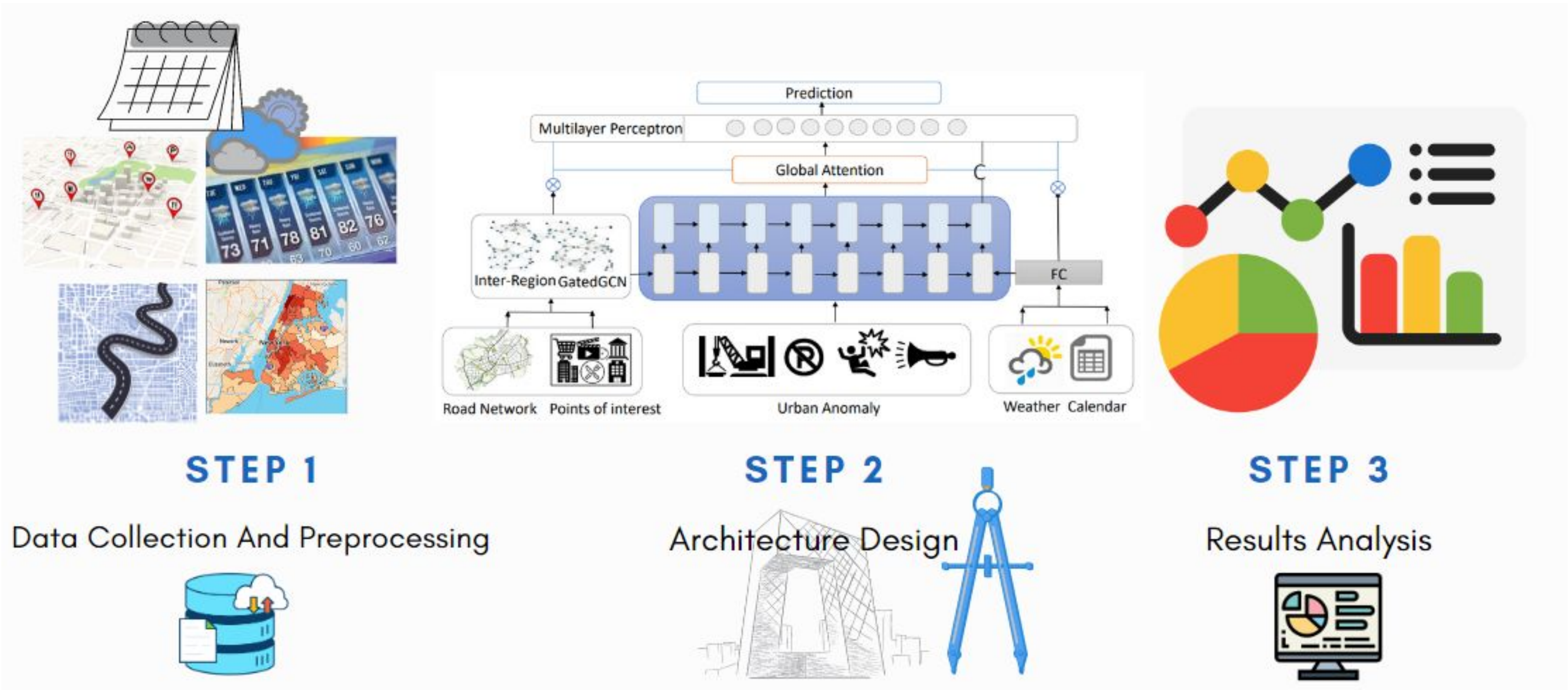
- In the training stage, a model is learned based on the **training data while in the testing stage**, the trained model is applied to produce the prediction.

- We briefly introduce the basics of machine learning in the hope that the readers can appreciate their **potential in solving traditionally challenging problems.**

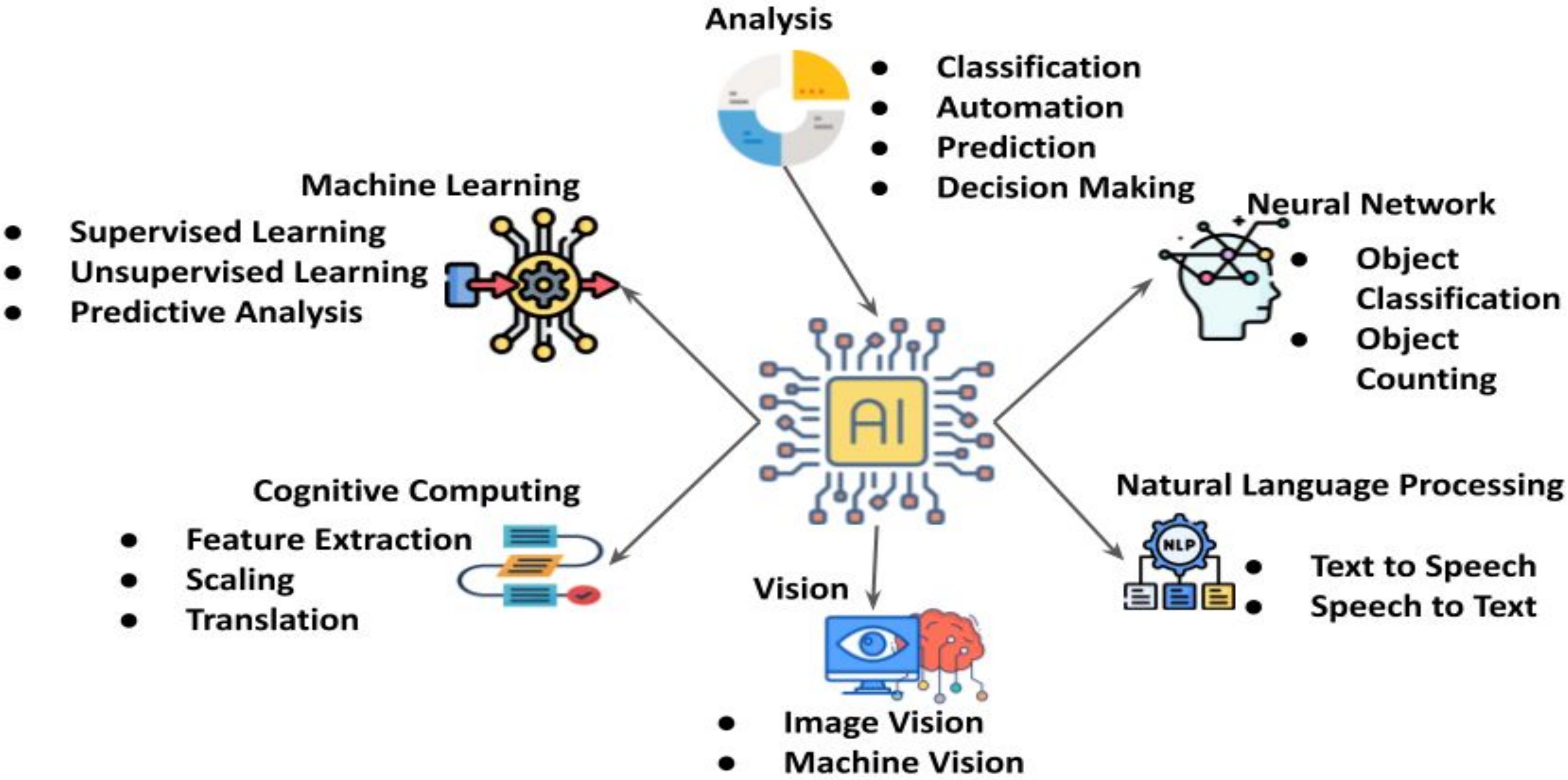
An illustration of deep neural networks with multiple layers



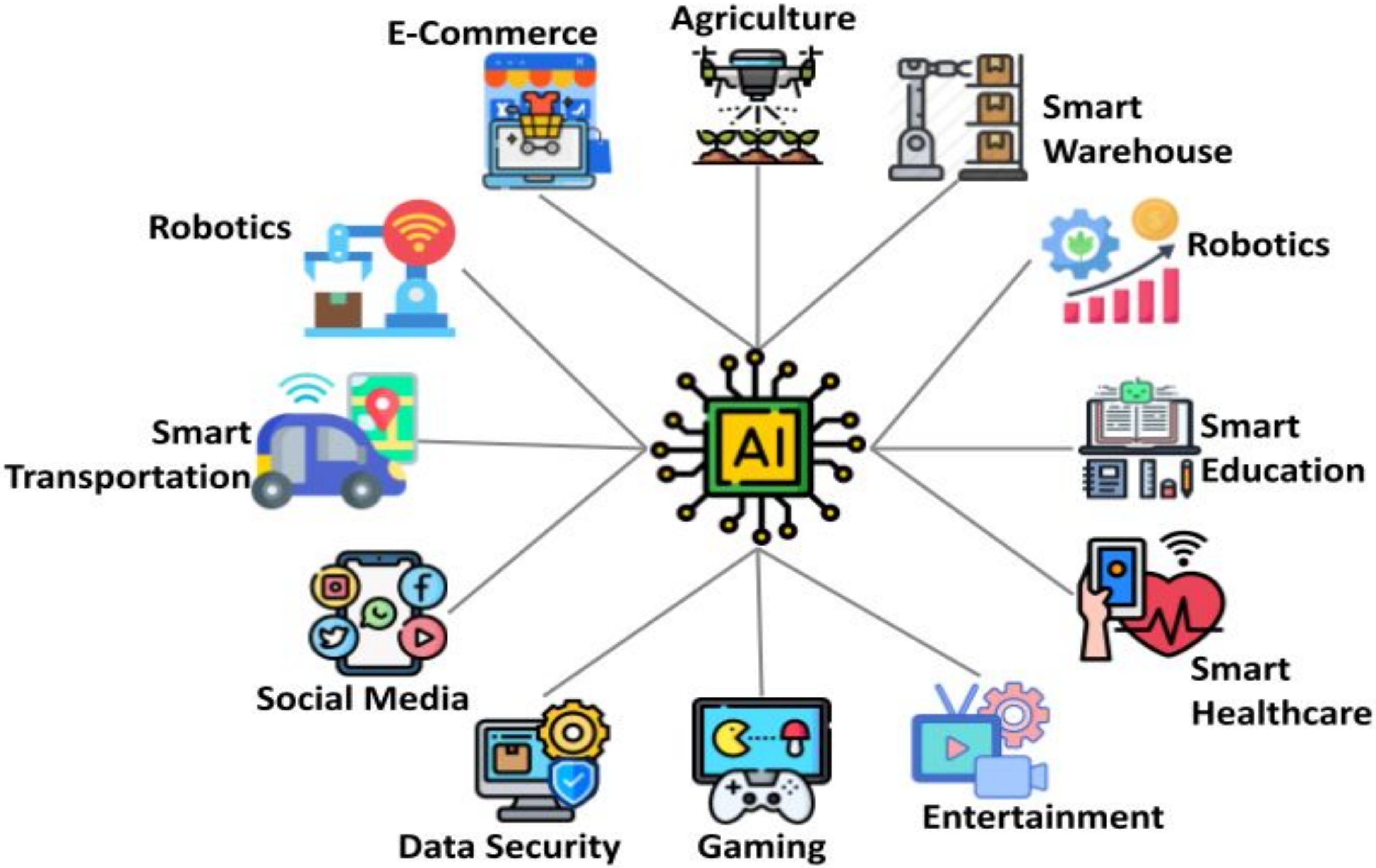
Methodology



AI Systems Components



Applications



Learning from Examples: Machine Learning

Up until now in this course:

- **Hand-craft algorithms** to make rational/optimal or at least good decisions.
  Examples: Search strategies, heuristics.

Issues

- Designer cannot anticipate all possible future situations.
- Designer may have examples but does not know how to program a solution.

Machine Learning

- Learning = Improve performance after making observations about the world.
  That is, learn what works and what doesn't.
- We learn a model that decides on the actions to take. This is called the "performance element."
- The goal is to get closer to optimal decisions. I.e., it is an optimization problem.

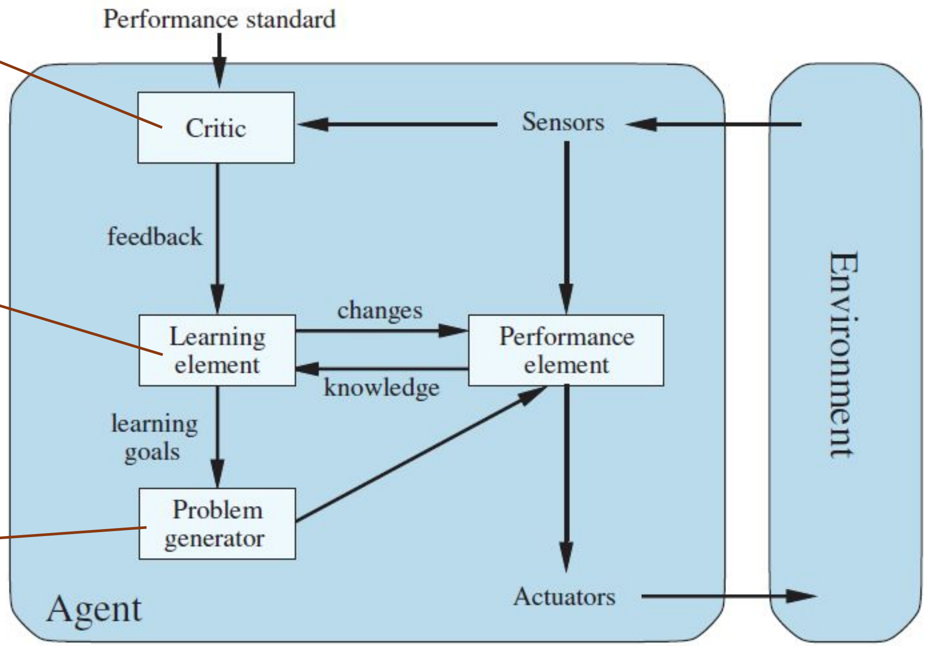
From Chapter 2: Agents that Learn

The **learning element** modifies the performance element to improve its performance.

Critic: How is the agent currently performing?

Learning Element: Update the performance element and changes how it selects actions.
E.g., adding rules, changing weights

Problem generators: Explore actions that lead to better information



Types of Using Machine Learning

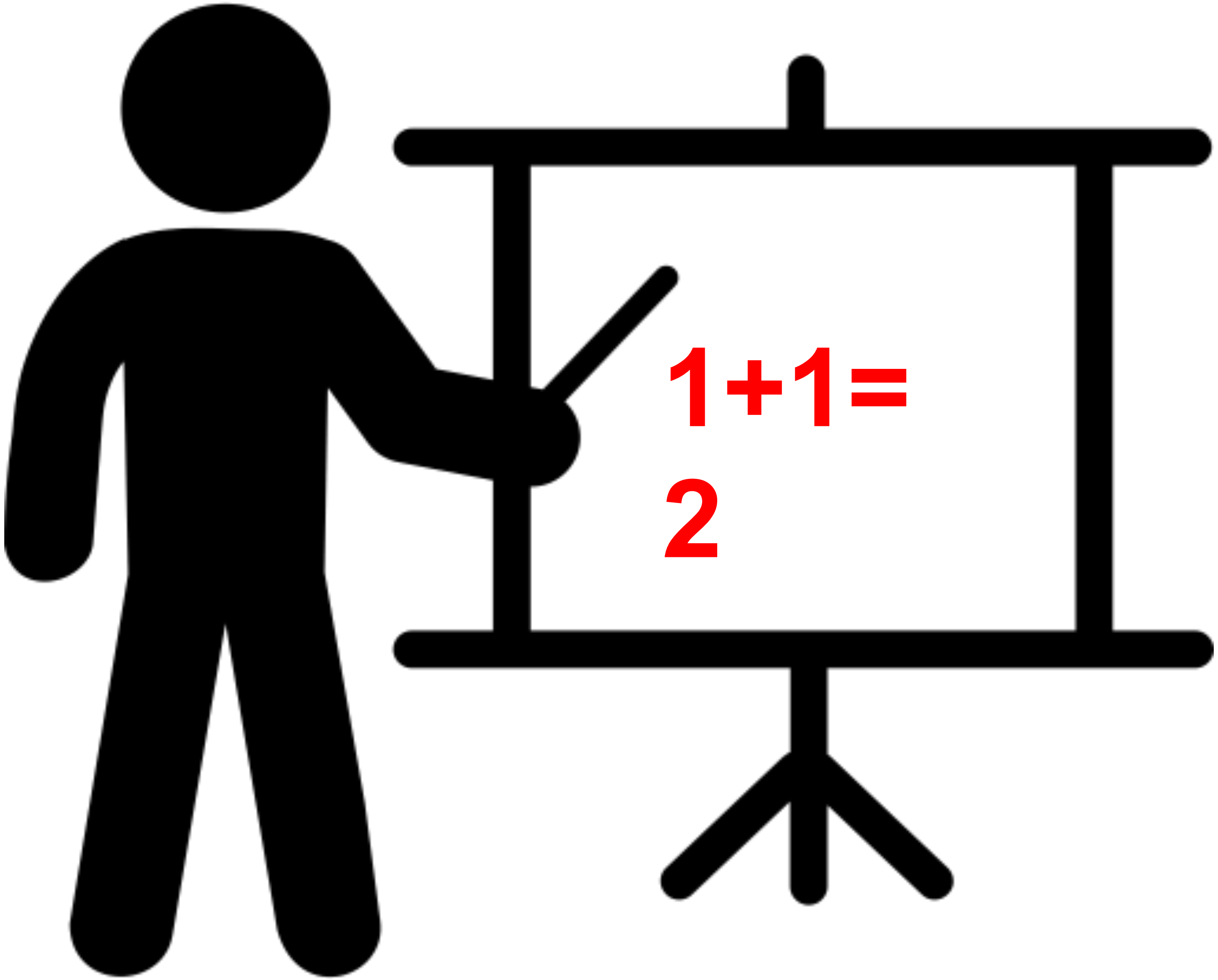
1. What **component** of the performance element is learned?
   E.g., how to select action, estimate the utility of a state, …

2. What **representation** (model) is used in the component?
   Linear regression, rules, trees, neural nets,…

3. What **feedback** is available for learning?
   - **Unsupervised Learning**: No feedback, just organize data (e.g., clustering, embedding)
   - **Supervised Learning**: Uses a data set with correct answers. Learn a function (model) to map an input (e.g., state) to an output (e.g., action or utility).
     Examples:
     - Use a naïve Bayesian classifier to distinguish between spam/no spam
     - Learn a playout policy to simulate games (current board -> good move)
   - **Reinforcement Learning**: Learn from rewards/punishment (e.g., winning a game) obtained via interaction with the environment over time.

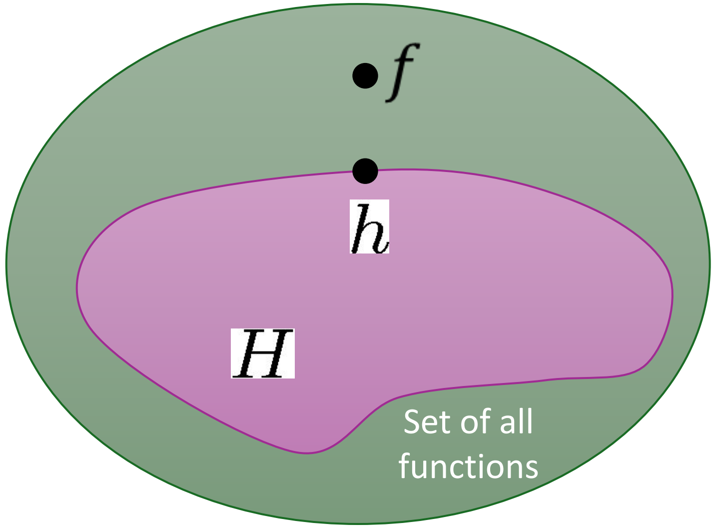
We focus here on supervised learning

Supervised Learning



Supervised Learning As Function Approximation

- Examples
  - We assume there exists a target function $y = f(x)$ that produces iid (independent and identically distributed) examples possibly with noise and errors.
  - Examples are observed input-output pairs $\mathrm{E} = (x_1, y_1), \ldots, (x_i, y_i), \ldots, (x_N, y_N)$, where x is a vectors called the feature vector.

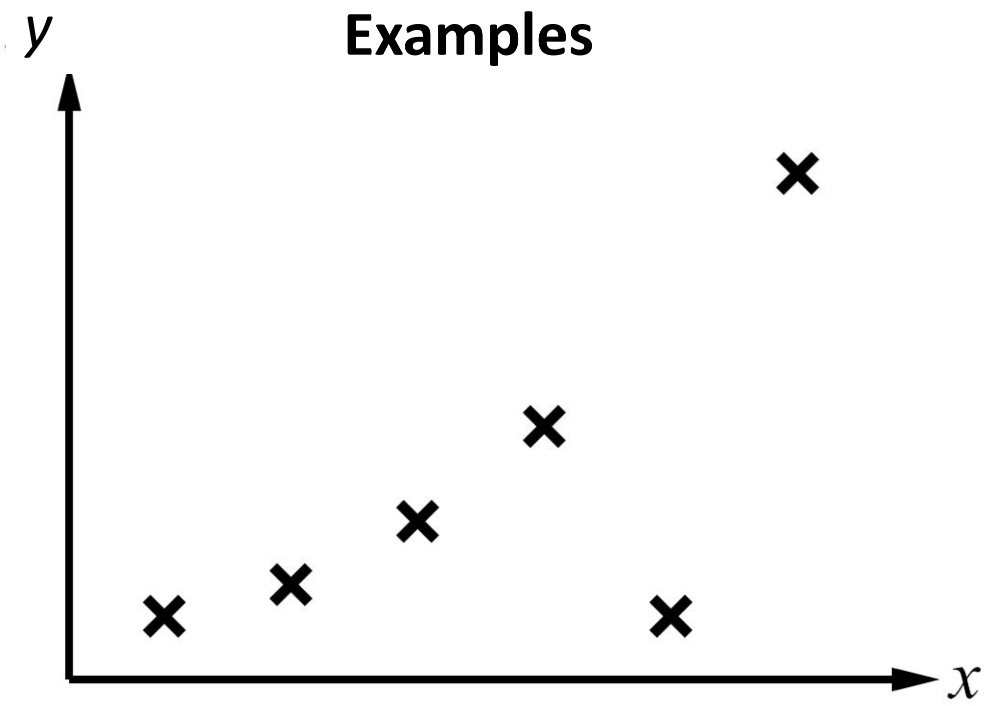


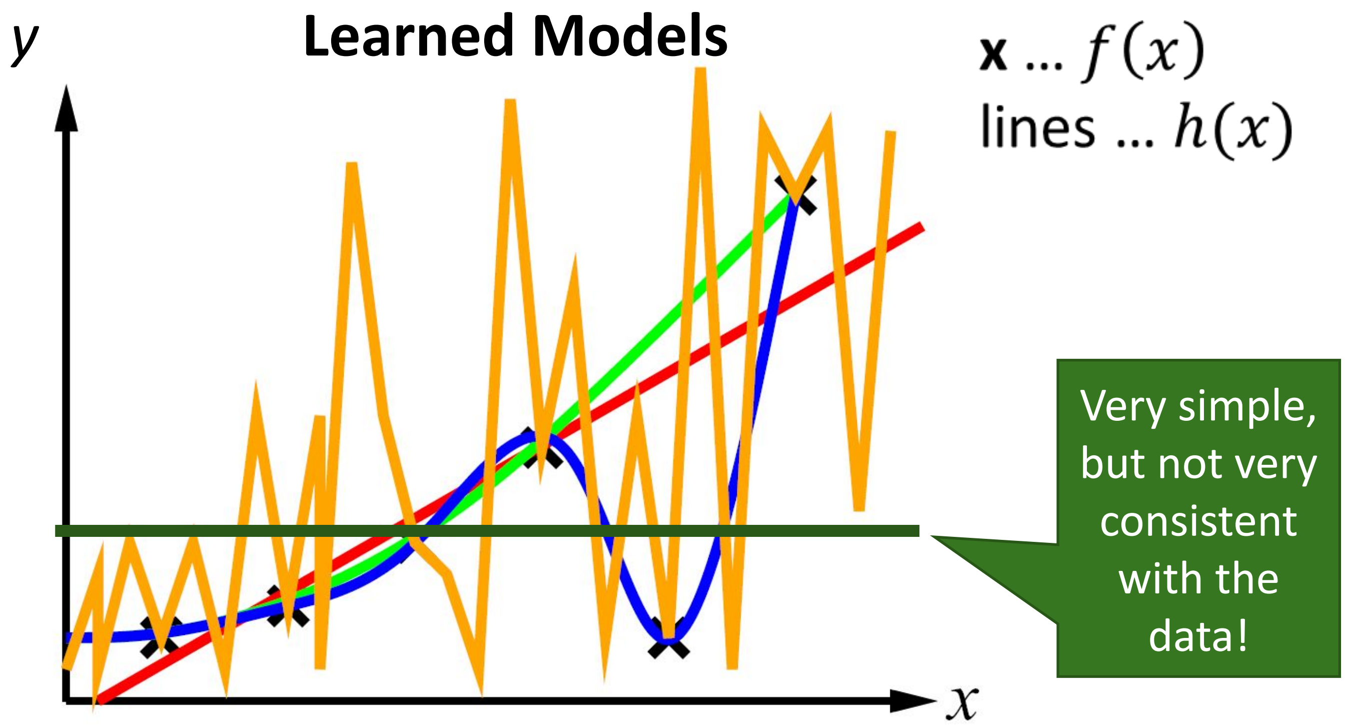
- Learning problem
  - Given a hypothesis space H of representable models.
  - Find a hypothesis $h \in H$ such that $\hat{y}_i = h(x_i) \approx y_i \; \forall i$
  - That is, we want to approximate f by h using E.

- Supervised learning includes
  - Classification (outputs = class labels). E.g., x is an email and $f(x)$ is spam / ham.
  - Regression (outputs = real numbers). E.g., x is a house and $f(x)$ is its selling price.

Consistency vs. Simplicity

- Example: Univariate curve fitting (regression, function approximation)





- **Consistency:** $h(x_i) \approx y_i$
- **Simplicity:** small number of model parameters

Measuring Consistency using Loss

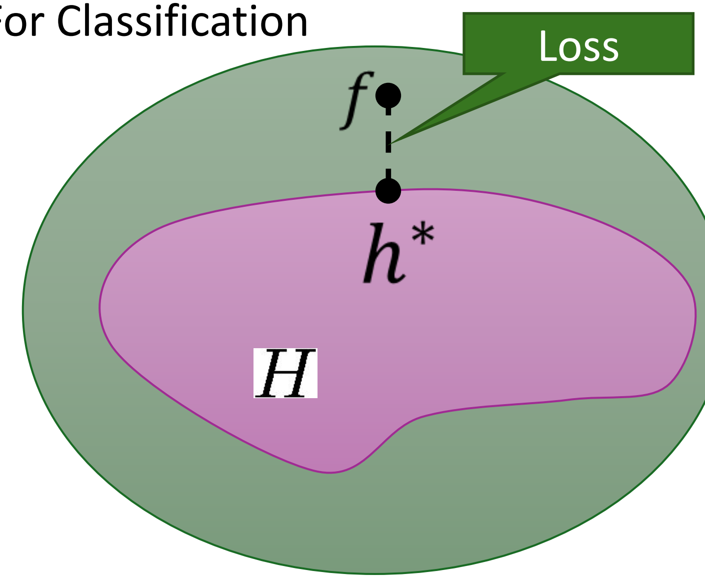
Goal of learning: Find a hypothesis that makes predictions that are consistent with the examples $\mathrm{E} = (x_1, y_1), \ldots, (x_i, y_i), \ldots, (x_N, y_N)$.

That is, $$\hat{y} = h(x) \approx y.$$

- **Measure mistakes:** Loss function $L(y, \hat{y}) = L(f(x), h(x))$
  - Absolute-value loss $L_1(y, \hat{y}) = |y - \hat{y}|$
  - Squared-error loss $L_2(y, \hat{y}) = (y - \hat{y})^2$
  - 0/1 loss $L_{0/1}(y, \hat{y}) = 0 \text{ if } y = \hat{y}, \text{else } 1$
  - Log loss, cross-entropy loss and many others…

For Regression

For Classification



Learning Consistent h by Minimizing the Loss

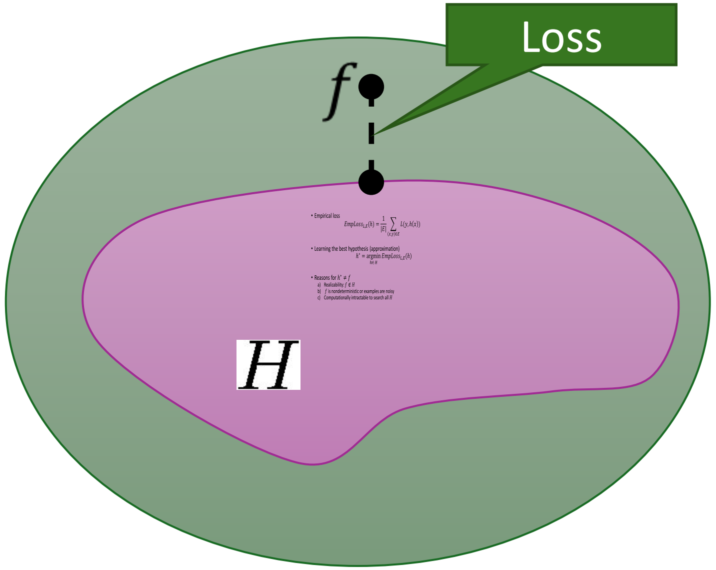
- Empirical loss

$$EmpLoss_{L,E}(h) = \frac{1}{|E|} \sum_{(x,y)\in E} L(y, h(x))$$

- Find the best hypothesis that minimizes the loss

$$h^* = \underset{h\in H}{\operatorname{argmin}}\, EmpLoss_{L,E}(h)$$

- Reasons for $h^* \neq f$
  a) Realizability: $f \notin H$
  b) f is nondeterministic or examples are noisy.
  c) It is computationally intractable to search all H, so we use a non-optimal heuristic.



The Most Consistent Classifier
The Bayes Classifier

For 0/1 loss, the empirical loss is minimized by the model that predicts for each x the most likely class y using MAP (Maximum a posteriori) estimates. This is called the Bayes classifier.

$$\mathrm{h}^*(\mathrm{x}) = \underset{y}{\operatorname{argmax}}\, P(Y = y \mid X = x) = \underset{y}{\operatorname{argmax}} \frac{P(x \mid y)\, P(y)}{P(x)} = \underset{y}{\operatorname{argmax}}\, P(x \mid y)\, P(y)$$

Optimality: The **Bayes classifier is optimal for 0/1 loss.** It is the most consistent classifier possible with the lowest possible error called the **Bayes error rate**. No better classifier is possible!

Issue: The classifier requires to learn $P(x \mid y)\, P(y) = P(x, y)$ from the examples.

- It **needs the complete joint probability** which requires in the general case a probability table with one entry for each possible value for the feature vector x.
- This is impractical (unless a simple Bayes network exists) and most classifiers try to approximate the Bayes classifier using a **simpler model** with fewer parameters.

Simplicity

Ease of use

- Simpler hypotheses have fewer model parameters to estimate and store.

Generalization: How well does the hypothesis perform on new data?

- We do not want the model to be too specific to the training examples (an issue called **overfitting**).
- Simpler models typically generalize better to new examples.

How to achieve simplicity?

a) **Model bias:** Restrict H to simpler models (e.g., assumptions like independence, only consider linear models).
b) **Feature selection:** use fewer variables from the feature vector x
c) **Regularization:** penalize model for its complexity (e.g., number of parameters)

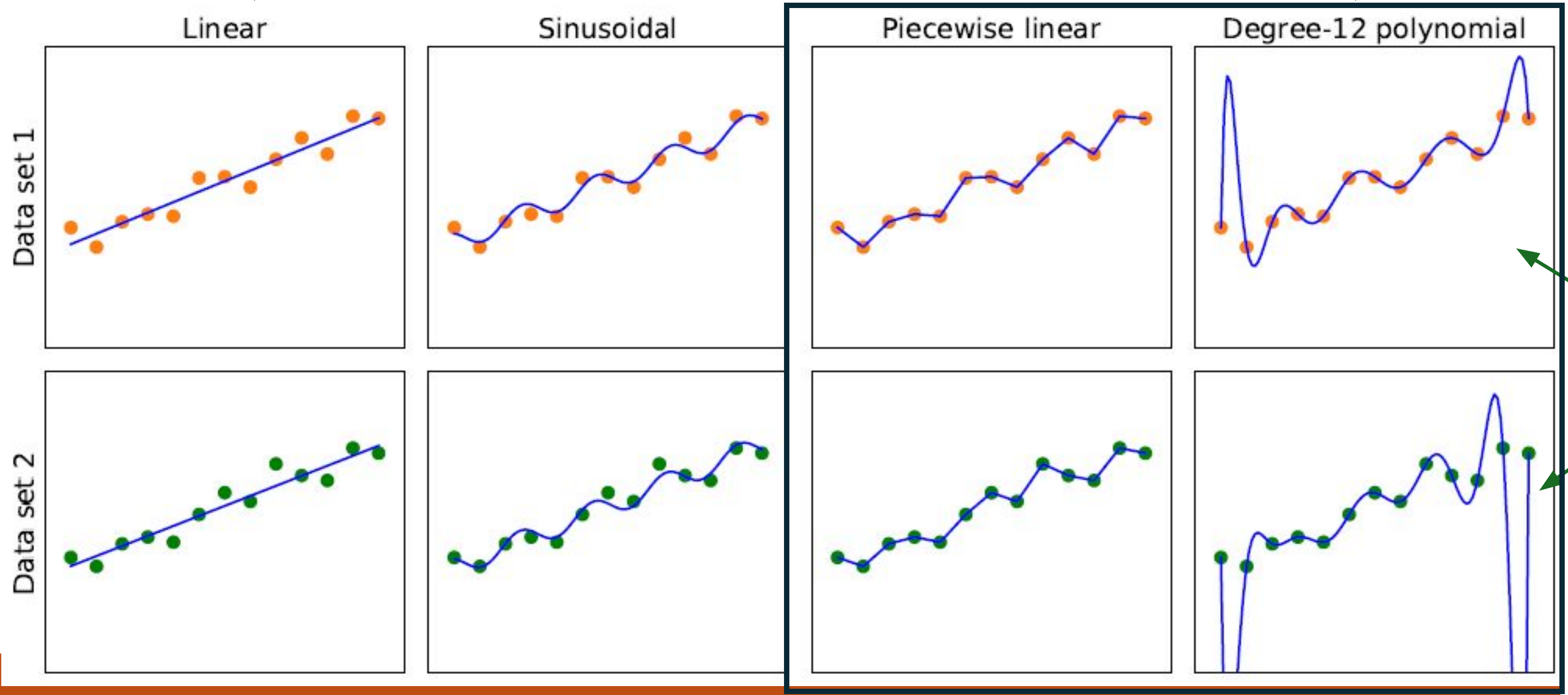
$$h^* = \underset{h \in H}{\operatorname{argmin}} \left[EmpLoss_{L,E}(h) + \lambda \underbrace{Complexity(h)}_{\text{Penalty term}} \right]$$

Model Selection: Bias vs. Variance

Overfitting

Simpler

More consistent



Points: Two samples from the same generating function f.

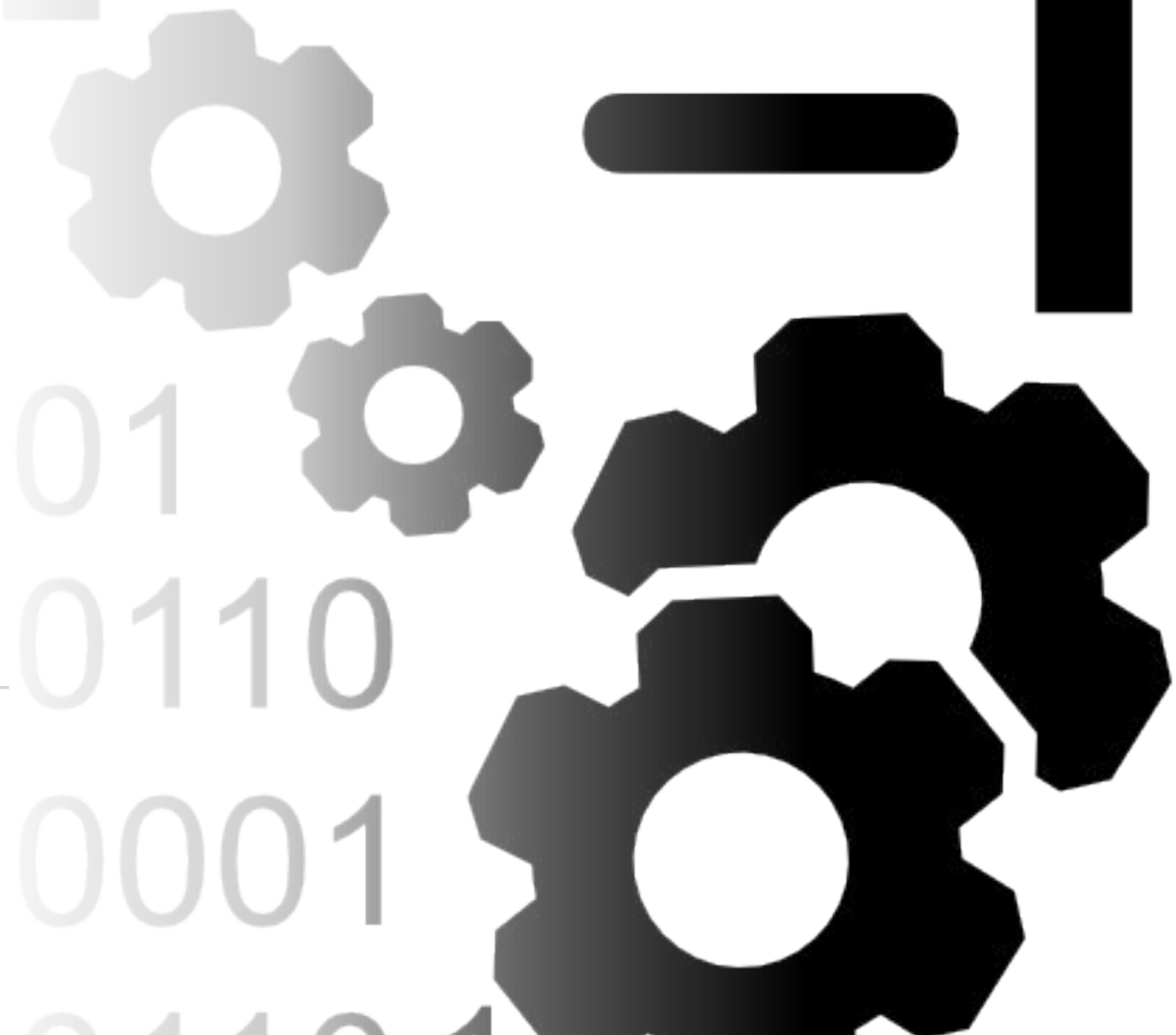
Lines: the learned model function h^*.

High **Bias**: restrictions by the model class Low

Low **Variance**: difference in the model due to slightly different data. high

This is a tradeoff

Data



The Dataset

Feature vector x
(Features, Variables, Attributes)

Class
Label y

Examples
(Instances,
Observation)

Example	Input Attributes										Output
	Alt	*Bar*	*Fri*	*Hun*	*Pat*	*Price*	*Rain*	*Res*	*Type*	*Est*	*WillWait*
$\mathbf{x}_1$	*Yes*	*No*	*No*	*Yes*	*Some*	*\$\$\$*	*No*	*Yes*	*French*	*0–10*	$y_1 = Yes$
$\mathbf{x}_2$	*Yes*	*No*	*No*	*Yes*	*Full*	*\$*	*No*	*No*	*Thai*	*30–60*	$y_2 = No$
$\mathbf{x}_3$	*No*	*Yes*	*No*	*No*	*Some*	*\$*	*No*	*No*	*Burger*	*0–10*	$y_3 = Yes$
$\mathbf{x}_4$	*Yes*	*No*	*Yes*	*Yes*	*Full*	*\$*	*Yes*	*No*	*Thai*	*10–30*	$y_4 = Yes$
$\mathbf{x}_5$	*Yes*	*No*	*Yes*	*No*	*Full*	*\$\$\$*	*No*	*Yes*	*French*	*>60*	$y_5 = No$
$\mathbf{x}_6$	*No*	*Yes*	*No*	*Yes*	*Some*	*\$\$*	*Yes*	*Yes*	*Italian*	*0–10*	$y_6 = Yes$
$\mathbf{x}_7$	*No*	*Yes*	*No*	*No*	*None*	*\$*	*Yes*	*No*	*Burger*	*0–10*	$y_7 = No$
$\mathbf{x}_8$	*No*	*No*	*No*	*Yes*	*Some*	*\$\$*	*Yes*	*Yes*	*Thai*	*0–10*	$y_8 = Yes$
$\mathbf{x}_9$	*No*	*Yes*	*Yes*	*No*	*Full*	*\$*	*Yes*	*No*	*Burger*	*>60*	$y_9 = No$
$\mathbf{x}_{10}$	*Yes*	*Yes*	*Yes*	*Yes*	*Full*	*\$\$\$*	*No*	*Yes*	*Italian*	*10–30*	$y_{10} = No$
$\mathbf{x}_{11}$	*No*	*No*	*No*	*No*	*None*	*\$*	*No*	*No*	*Thai*	*0–10*	$y_{11} = No$
$\mathbf{x}_{12}$	*Yes*	*Yes*	*Yes*	*Yes*	*Full*	*\$*	*No*	*No*	*Burger*	*30–60*	$y_{12} = Yes$

Alternative

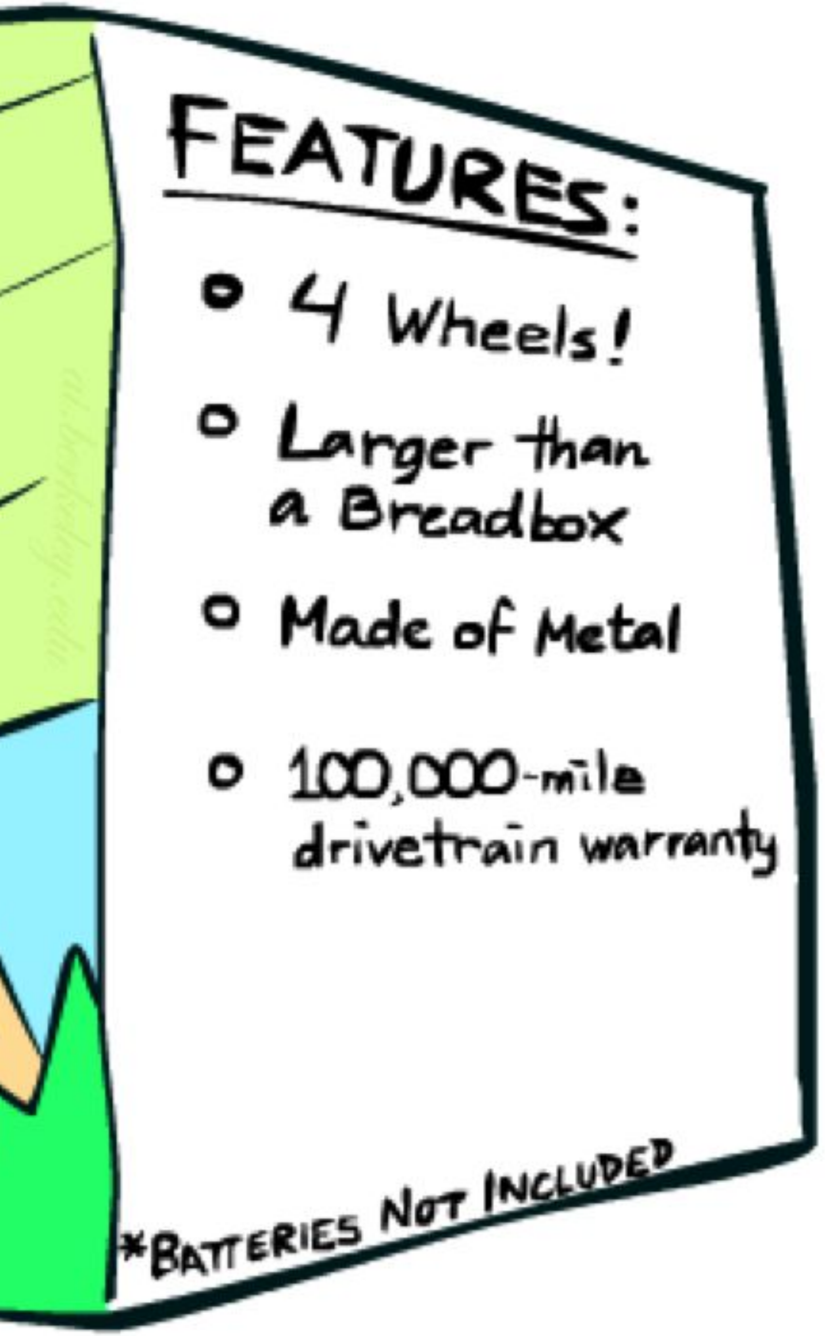
Hungry

Patrons

Reservation

Wait time

Find a hypothesis (called “model”) to predict the class given the features.



Feature Engineering

- Add information sources as new variables to the model.
- Add derived features that help the classifier (e.g., $x_1 x_2$, x_1^2).
- Embedding: E.g., convert words to vectors where vector similarity between vectors reflects semantic similarity.

- Example for Spam detection: In addition to words
  - Have you emailed the sender before?
  - Have 1000+ other people just gotten the same email?
  - Is the email in ALL CAPS?
- **Feature Selection**: Which features should be used in the model is a model selection problem (choose between models with different features).

Data in AI

- Data in AI can come from many sources
  - **Observation**: Record video of a task being performed.
  - **Existing Data**: Download documents from the internet to train Large Language Models.
  - **Simulation**: E.g., simulated games using a playout strategy.
  - **Expert feedback** on how well a task was performed.

Training and Testing

Training a Model

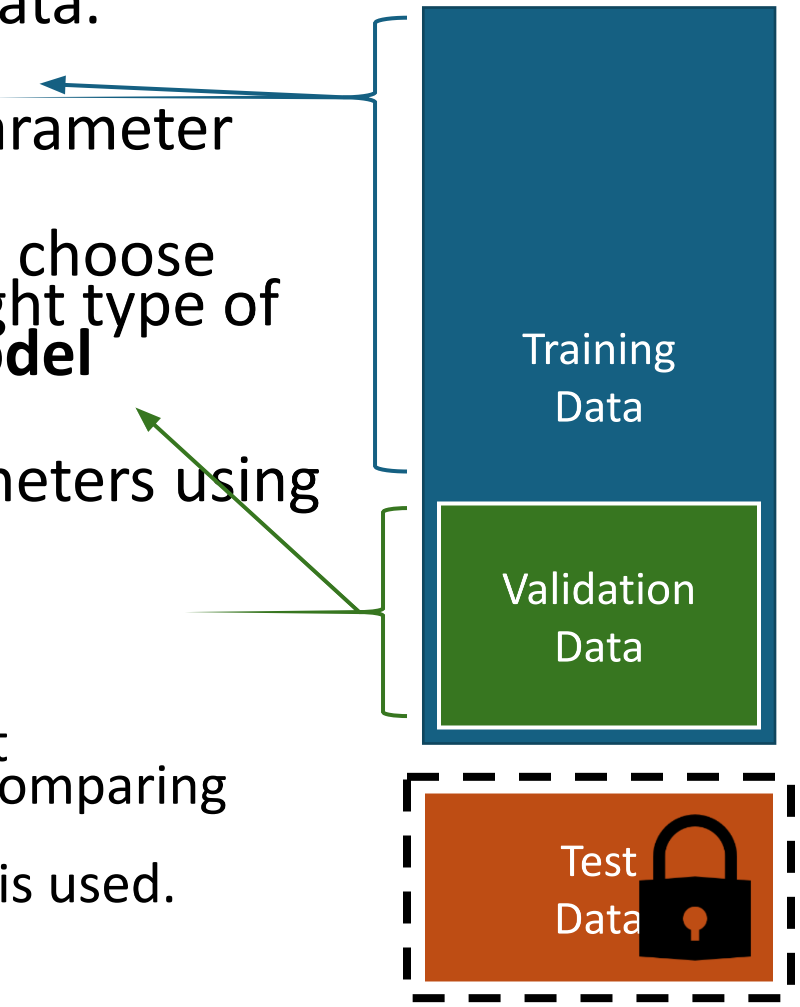
- Models are "trained" (learned) on **the training data.** This involved estimating:

  1. **Model parameters** (the model): E.g., probabilities, weights, factors.
  2. **Hyperparameters**: Many learning algorithms have choices for learning rate, regularization λ, maximal decision tree depth, selected features,... The algorithm tries to optimizes the model parameters given user-specified hyperparameters.

- We need to tune the hyperparameters! This is a type of model selection.

Hyperparameter Tuning/Model Selection

1. Hold a validation data set back from the training data.
2. **Learn models** using the training set with different hyperparameters. Often a grid of possible hyperparameter combinations or some greedy search is used.
3. **Evaluate the models** using the validation data and choose the model with the best accuracy. Selecting the right type of model, hyperparameters and features is called **model selection**.
4. Learn the final model with the chosen hyperparameters using all training (including validation data).



- Notes:
  - The validation set was not used for training with different hyperparameters, so we get generalization accuracy for comparing different hyperparameter settings.
  - If no model selection is necessary, then no validation set is used.

Model Evaluation (Testing)

The model was trained on the training examples E. We want to test how well the model will perform on new examples T (i.e., how well it **generalizes to new data**). We use held back test data.

- **Testing loss**: Calculate the empirical loss for predictions on a testing data set T that is different from the data used for training.

$$EmpLoss_{L,T}(h) = \frac{1}{|T|} \sum_{(x,y)\in T} L(y, h(x))$$

- For classification we often use the **accuracy** measure, the proportion of correctly classified test examples.

$$accuracy(h,T) = \frac{1}{|T|} \sum_{(x,y)\in T} [h(x) = y] = 1 - \ EmpLoss_{L_{0/1},T}(h)$$

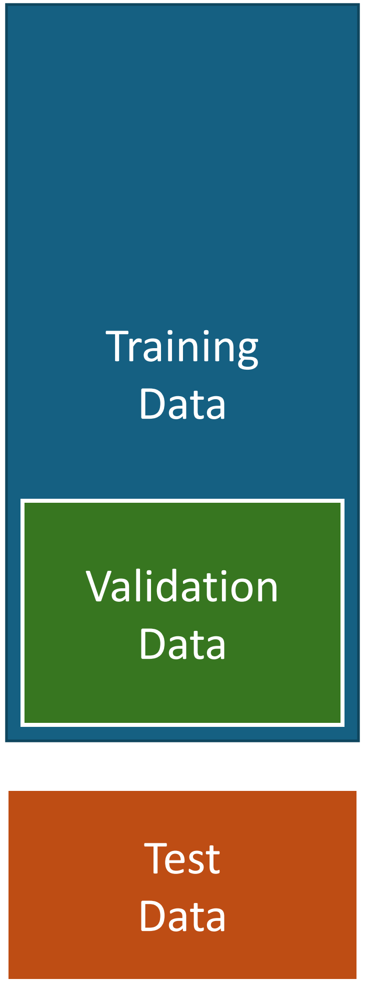
$[c]$ is an indicator function returning 1 if $c \ = \ True$ and otherwise 0

How to Split the Dataset

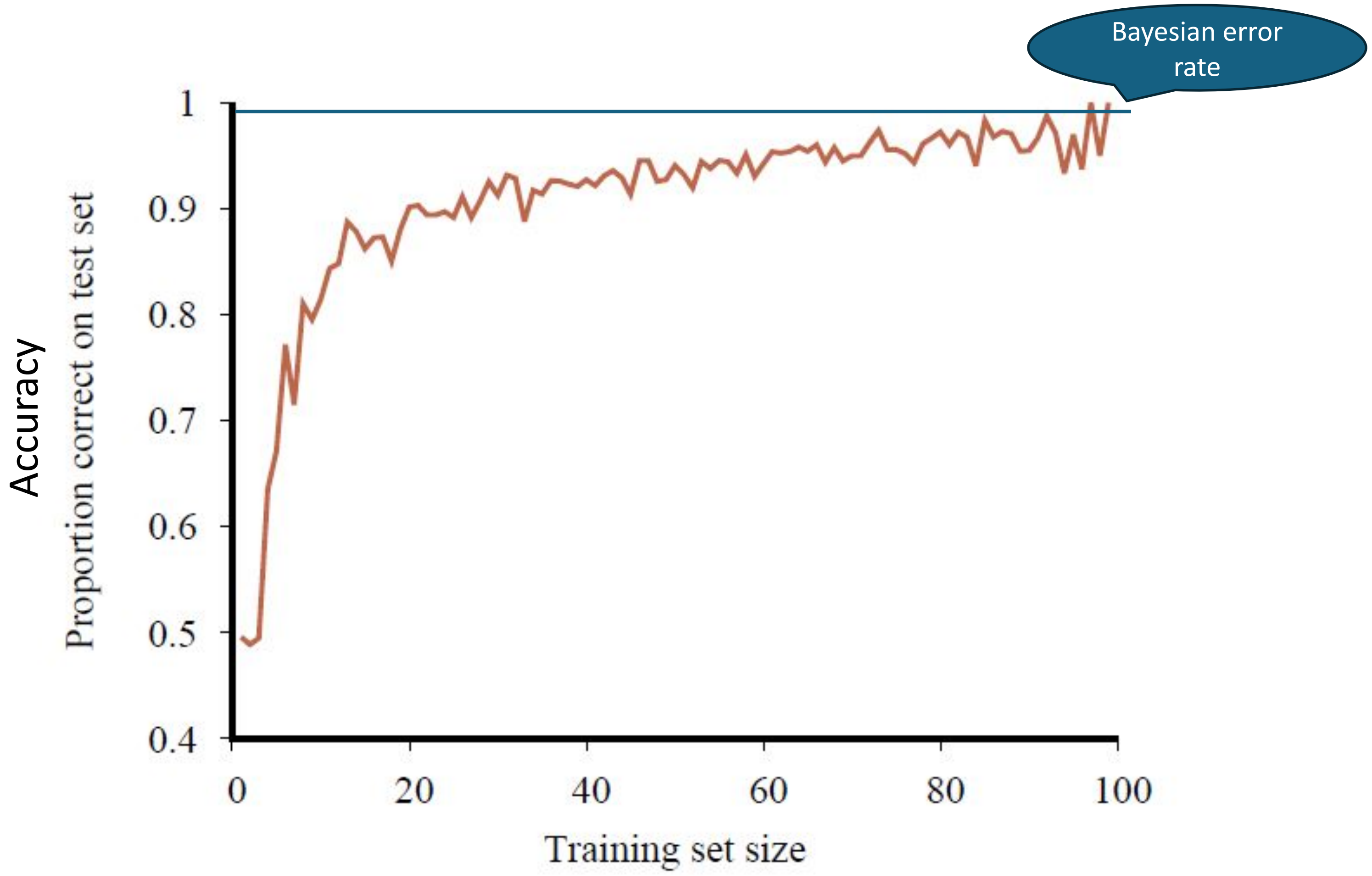
- **Random splits:** Split the data randomly in, e.g., 60% training, 20% validation, and 20% testing.

- **Stratified splits:** Like random splits, but balance classes or other properties of the examples.

- **k-fold cross validation:** Use training & validation data better
  - Split the training & validation data randomly into k folds.
  - For each of k rounds, hold one fold back for testing and use the remaining $k-1$ folds for training.
  - Use the average error/accuracy as a better estimate.
  - Some algorithms/tools do this internally.



Learning Curve:
The Effect the Training Data Size



Accuracy increases when the amount of available training data increases.

More data is better!

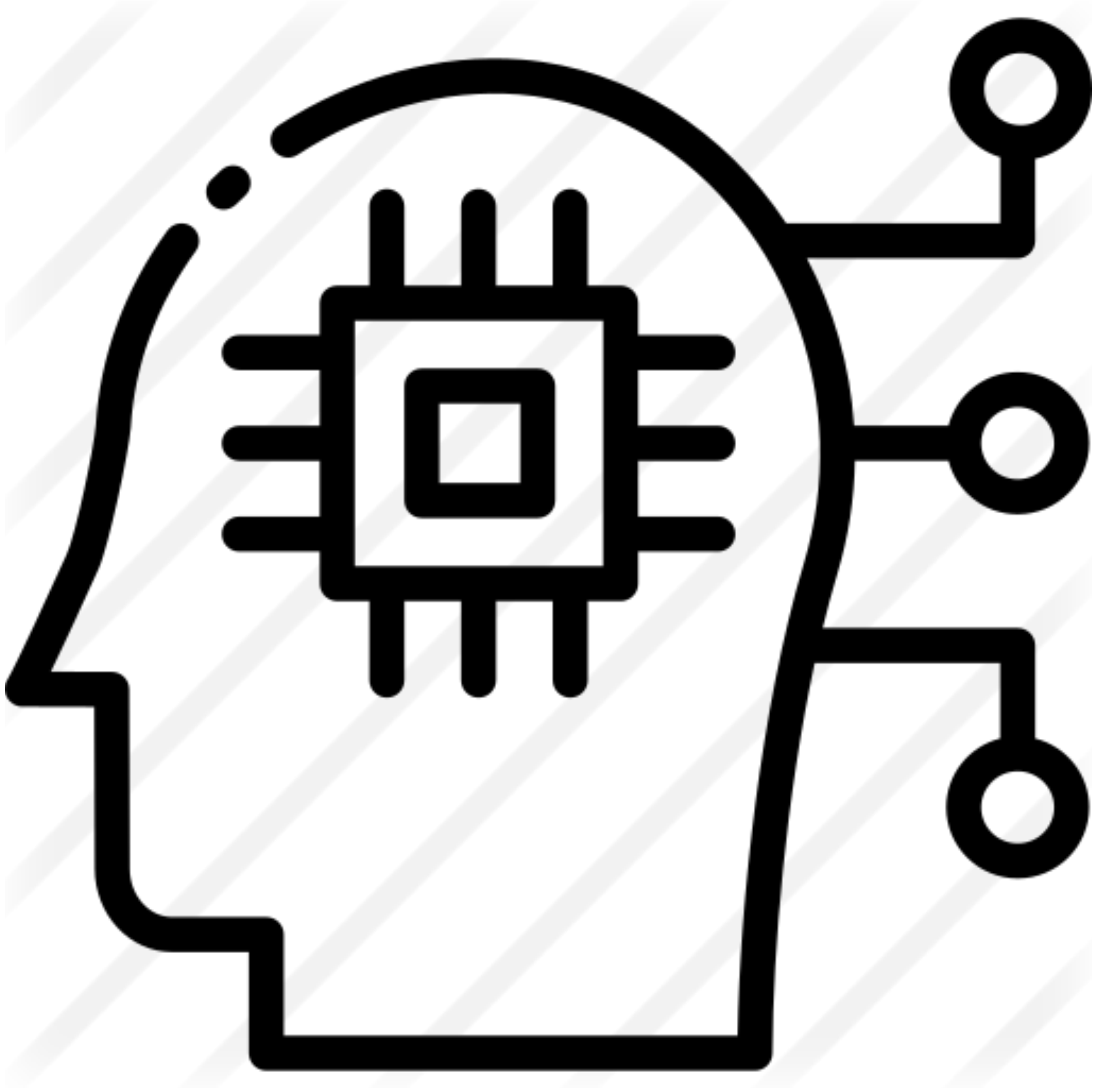
At some point the learning curve flattens out and more data does not contribute much!

Comparing to a Baselines

- First step: get a **baseline**
  - Baselines are very simple straw man model.
  - Helps to determine how hard the task is.
  - Helps to find out what a good accuracy is.

- **Weak baseline**: The most frequent label classifier
  - Gives all test instances whatever label was most common in the training set.
    - Example: For spam filtering, give every message the label “ham.”
  - Accuracy might be very high if the problem is skewed (called class imbalance).
    - Example: If calling everything “ham” gets already 66% right, so a classifier that gets 70% isn’t very good…

- **Strong baseline**: For research, we typically compare to previous published state-of-the-art as a baseline.



Types of ML Models

Regression: Predict a number

Classification: Predict a label

Regression: Linear Regression

Model: $h_{\boldsymbol{w}}(\boldsymbol{x}_j) = w_o + w_1 x_{j,1} + \cdots + w_n x_{j,n} = \sum_i w_i x_{j,i} = \boldsymbol{w}^T \boldsymbol{x}_j$

Empirical Loss: $L(\boldsymbol{w}) = \|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2$

Squared error loss over the whole data matrix $\boldsymbol{X}$

Gradient: $\nabla L(\boldsymbol{w}) = 2X^T(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y})$

The gradient is a vector of partial derivatives

$$\nabla L(\boldsymbol{w}) = \left[\frac{\partial L}{\partial w_1}(\boldsymbol{w}), \frac{\partial L}{\partial w_2}(\boldsymbol{w}), \dots, \frac{\partial L}{\partial w_n}(\boldsymbol{w})\right]^T$$

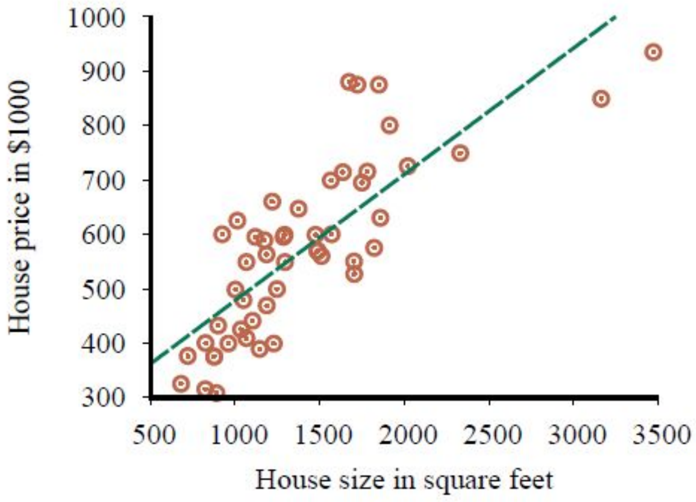
Find: $\nabla L(\boldsymbol{w}) = 0$

Gradient descend:

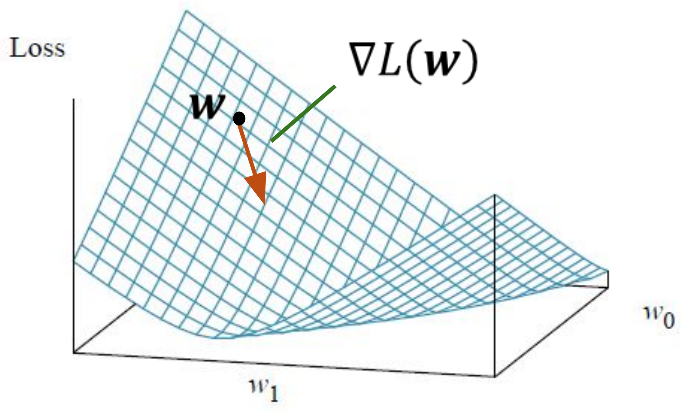
$$\boldsymbol{w} = \boldsymbol{w} - \alpha \nabla L(\boldsymbol{w})$$

Analytical solution:

$$\boldsymbol{w}^* = \underbrace{(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T}_{\text{Pseudo inverse}}\boldsymbol{y}$$



(a)



(b)

Naïve Bayes Classifier

Bayes Classifier
$$h^*(x) = \operatorname*{argmax}_y P(Y = y \mid X = x)$$

- Approximates a Bayes classifier with the **naïve independence assumption** that all n features are conditional independent given the class.

$$h(x) = \operatorname*{argmax}_y \; P(y) \prod_{i=1}^{n} P(x_i \mid y)$$

The $P(y)$s and the $P(x_i \mid y)$s are estimated from the data by counting.

- Gaussian Naïve Bayes Classifiers extend the approach to continuous features by assuming the feature follows a normal distribution depending on the class:

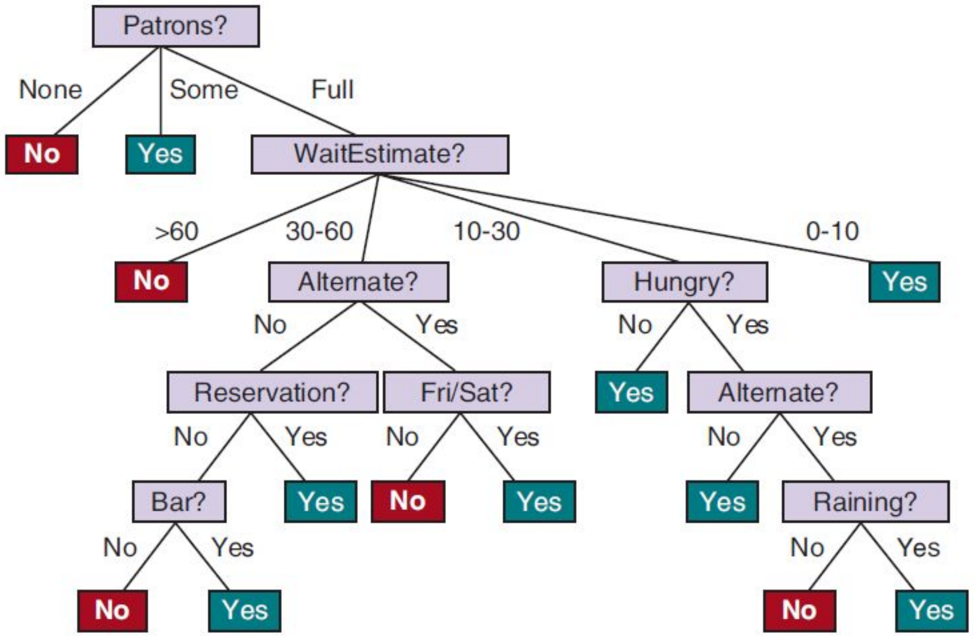
$$P(x_i \mid y) \sim N(\mu_y, \sigma_y)$$

The parameters for the normal distribution $N(\mu_y, \sigma_y)$ are estimated from data.

- A **sequence of decisions** represented as a tree.
- Many implementations that differ by
  - How to select features to split?
  - When to stop splitting?
  - Is the tree pruned?
- Approximates a Bayesian classifier by
$$h(x) = \underset{y}{\operatorname{argmax}}\ P(Y = y \mid \text{leafNodeMatching}(x))$$

Decision Trees

Example	Input Attributes										Output
	Alt	*Bar*	*Fri*	*Hun*	*Pat*	*Price*	*Rain*	*Res*	*Type*	*Est*	*WillWait*
$\mathbf{x}_1$	*Yes*	*No*	*No*	*Yes*	*Some*	*\$\$\$*	*No*	*Yes*	*French*	*0–10*	$y_1 = Yes$
$\mathbf{x}_2$	*Yes*	*No*	*No*	*Yes*	*Full*	*\$*	*No*	*No*	*Thai*	*30–60*	$y_2 = No$
$\mathbf{x}_3$	*No*	*Yes*	*No*	*No*	*Some*	*\$*	*No*	*No*	*Burger*	*0–10*	$y_3 = Yes$
$\mathbf{x}_4$	*Yes*	*No*	*Yes*	*Yes*	*Full*	*\$*	*Yes*	*No*	*Thai*	*10–30*	$y_4 = Yes$
$\mathbf{x}_5$	*Yes*	*No*	*Yes*	*No*	*Full*	*\$\$\$*	*No*	*Yes*	*French*	*>60*	$y_5 = No$
$\mathbf{x}_6$	*No*	*Yes*	*No*	*Yes*	*Some*	*\$\$*	*Yes*	*Yes*	*Italian*	*0–10*	$y_6 = Yes$
$\mathbf{x}_7$	*No*	*Yes*	*No*	*No*	*None*	*\$*	*Yes*	*No*	*Burger*	*0–10*	$y_7 = No$
$\mathbf{x}_8$	*No*	*No*	*No*	*Yes*	*Some*	*\$\$*	*Yes*	*Yes*	*Thai*	*0–10*	$y_8 = Yes$
$\mathbf{x}_9$	*No*	*Yes*	*Yes*	*No*	*Full*	*\$*	*Yes*	*No*	*Burger*	*>60*	$y_9 = No$
$\mathbf{x}_{10}$	*Yes*	*Yes*	*Yes*	*Yes*	*Full*	*\$\$\$*	*No*	*Yes*	*Italian*	*10–30*	$y_{10} = No$
$\mathbf{x}_{11}$	*No*	*No*	*No*	*No*	*None*	*\$*	*No*	*No*	*Thai*	*0–10*	$y_{11} = No$
$\mathbf{x}_{12}$	*Yes*	*Yes*	*Yes*	*Yes*	*Full*	*\$*	*No*	*No*	*Burger*	*30–60*	$y_{12} = Yes$

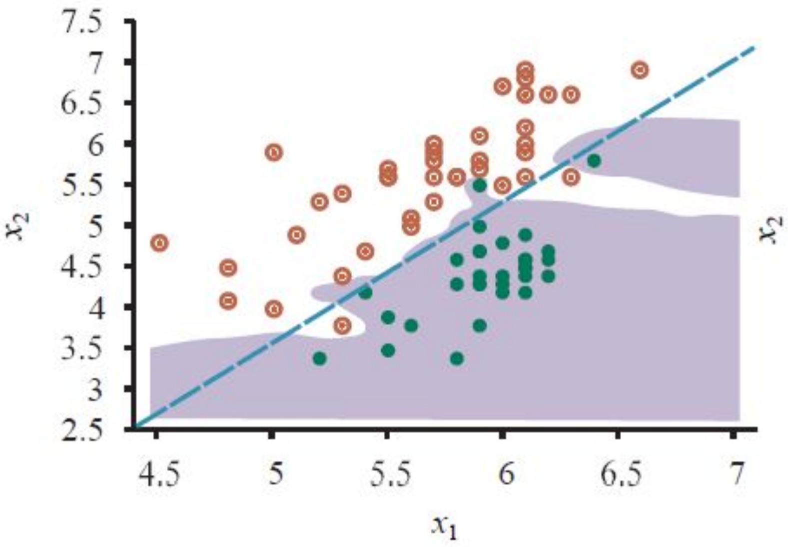


- A **sequence of decisions** represented as a tree.
- Many implementations that differ by
  - How to select features to split?
  - When to stop splitting?
  - Is the tree pruned?
- Approximates a Bayesian classifier by

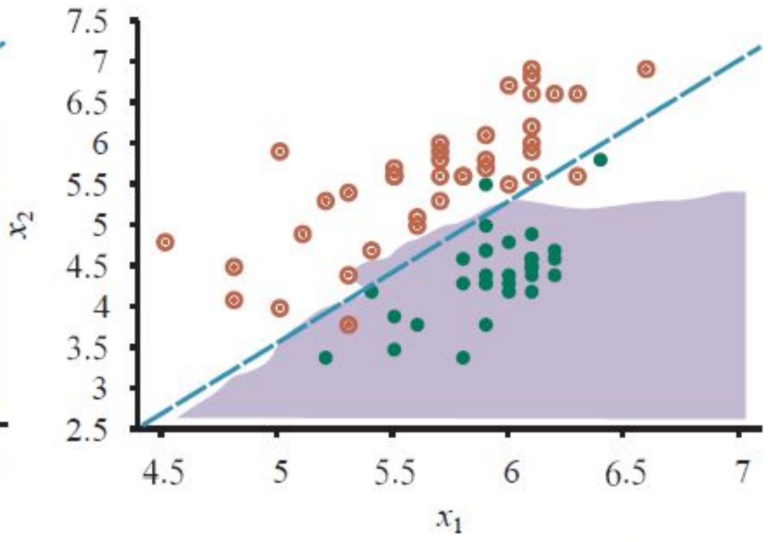
$$h(x) = \underset{y}{\operatorname{argmax}}\ P(Y = y \mid \text{leafNodeMatching}(x))$$

K-Nearest Neighbors Classifier

Bayes Classifier
$$h^*(x) = \underset{y}{\operatorname{argmax}}\, P(Y = y \mid X = x)$$



$(k = 1)$

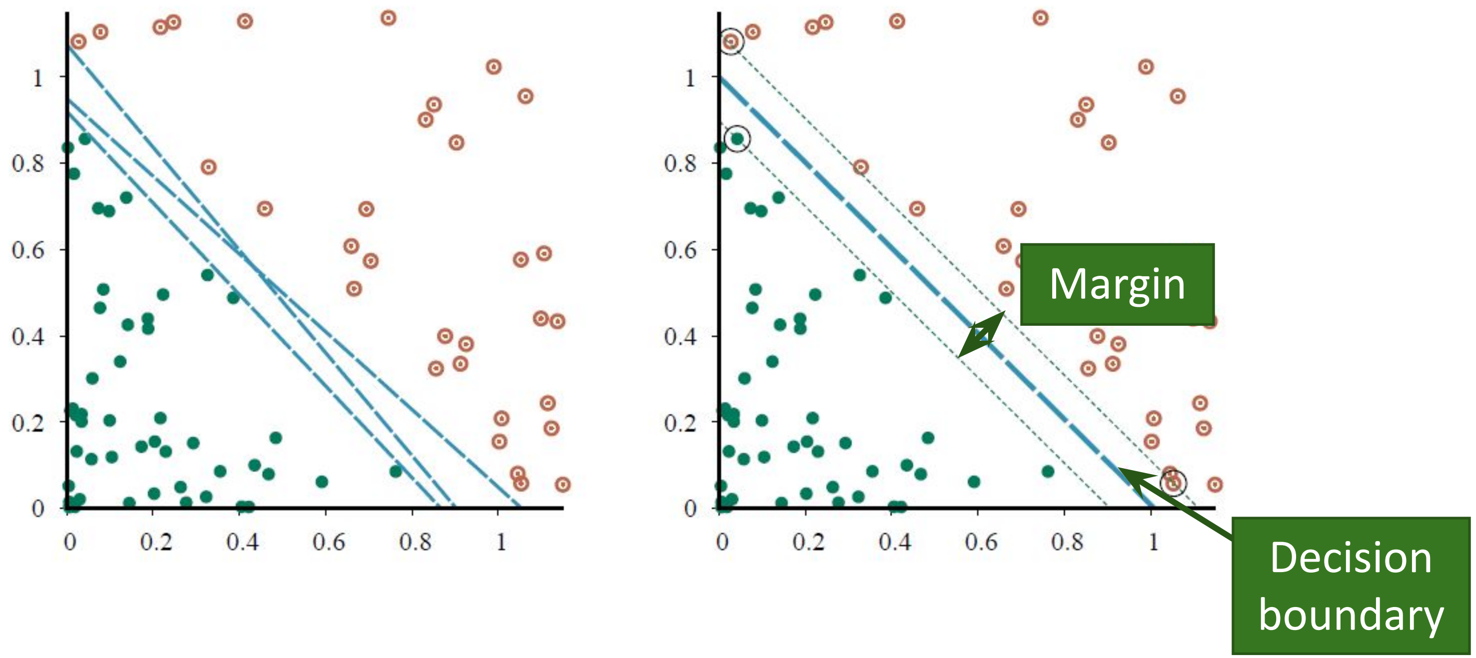


$(k = 5)$

- Class is predicted by looking at the majority in the set of the k nearest **neighbors**. k is a hyperparameter. Larger k smooth the decision boundary.
- Neighbors are found using a distance measure (e.g., Euclidean distance between points).
- Approximates a Bayesian classifier by

$$h(x) = \underset{y}{\operatorname{argmax}}\, P(Y = y \mid \text{neighborhood}(x))$$

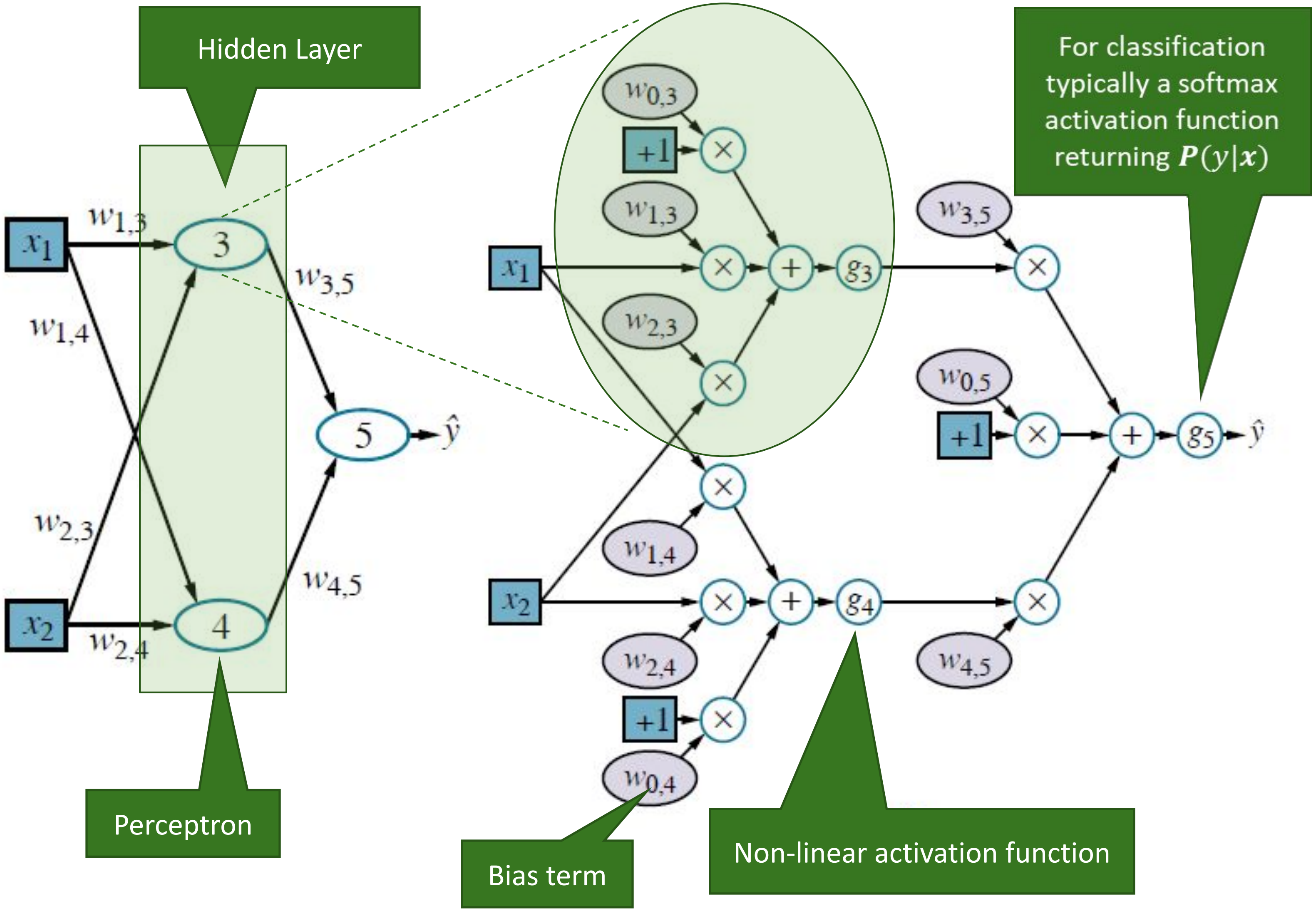
Support Vector Machine (SVM)



- Linear classifier that finds **the maximum margin separator** using only the points that are "support vectors" and quadratic optimization.
- The kernel trick can be used to learn non-linear decision boundaries.

Artificial Neural Networks/Deep Learning

Network Topology **Computational graph**



- Represent
  $\hat{y} = h(x) = g_2(\boldsymbol{W}_2\, g_1(\boldsymbol{W}_1 \boldsymbol{x}))$
  as a network of weighted sums with non-linear **activation functions** g (e.g., logistic, ReLU).
- Learn weight matrices $\boldsymbol{W}$ from examples using **backpropagation** of prediction errors $\mathrm{L}(\hat{y}, y)$ (gradient descend).
- ANNs are **universal approximators**. Large networks can approximate any function (no bias). **Regularization** is typically needed to avoid overfitting.
- **Deep learning** adds more hidden layers and layer types (e.g., convolution layers) for better learning.

Other Popular Models and Methods

Many other models exist

- **Generalized linear model (GLM):** This important model family includes **linear regression** and the classification method **logistic regression.**

Often used methods

- **Regularization:** enforce simplicity and reduces overfitting by using a penalty for complexity.
- **Kernel trick:** Let a linear classifier learn non-linear decision boundaries (= a linear boundary in a high dimensional space).
- **Ensemble Learning:** Use many models and combine the results (e.g., random forest, boosting).
- **Embedding and Dimensionality Reduction:** Learn how to represent data in a simpler way (e.g., PCA, text embeddings).

Some Use Cases of ML for Intelligent Agents

Learn Actions

- Directly learn the best action from examples.
- This model can also be used as a **playout policy** for Monte Carlo tree search with data from self-play.

Learn Heuristics

- Learn evaluation functions for states.
- Can learn a **heuristic** for minimax search from examples.

Perception

- **Natural language processing:** Use deep learning / word embeddings / language models to understand concepts, translate between languages, or generate text.
- Speech recognition: Identify the most likely sequence of words.
- Vision: Object recognition in images/videos. Generate images/video.

Compressing Tables

- Neural networks can be used as a compact representation of tables that do not fit in memory. E.g.,
  - Joint and conditional probability tables
  - State utility tables
- The tables can be learned form data.

Bottom line: Learning a function is often more effective than hard-coding it
However, we do not always know how it performs in very rare cases!